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Robust tightly coupled MLS–ALS fusion with 2D and 3D LiDAR using local geometric uncertainty for accurate 3D mapping

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ABSTRACT

Accurate, high-resolution 3D maps are essential for applications such as forestry, infrastructure monitoring, and autonomous navigation. Airborne laser scanning (ALS) provides large-scale coverage but suffers from canopy occlusions, whereas mobile laser scanning (MLS) captures detailed ground-level structure but is susceptible to drift when Global Navigation Satellite System-Inertial Navigation System (GNSS-INS) measurements degrade. Although ALS–MLS fusion is promising, most existing approaches rely on loosely coupled pipelines or feature-based alignment methods that require manual parameter tuning, depend on feature quality, and are not well-suited for real-time operation. On these mapping platforms, 3D LiDAR provides full motion observability but often with lower range accuracy and sparser sampling. In contrast, certain 2D LiDAR sensors offer higher range precision and point density but lack one degree of freedom, limiting direct 3D reconstruction. In this study, we propose a tightly coupled multi-sensor fusion framework that integrates IMU, 3D LiDAR, 2D LiDAR, GNSS-INS derived $SE(3)$ poses, and prior ALS maps for online localisation and mapping. The system is built on a LiDAR-inertial backbone, where 2D LiDAR measurements complement 3D LiDAR with higher point density and range precision. GNSS-INS or ALS maps provide global priors for MLS drift correction. All measurements are jointly optimised in a single-state maximum a posteriori formulation using robust, data-adaptive geometric uncertainty weighting based on local planarity. Sensor contributions are combined through accumulated Hessians and gradients with an adaptive self-balancing mechanism that removes manual parameter tuning. Experiments over 5 km of forest roads demonstrate high localisation accuracy compared to the reference baseline. Using dense ALS priors, the system achieves ≈ 0.04 m absolute error and ≈ 0.17% relative error, while sparse ALS priors yield ≈ 0.18 m and ≈ 0.39%, respectively. The source code of the proposed method is publicly available.[1]

1. Introduction

Accurate, high-resolution 3D maps are essential for forestry, road monitoring, urban planning, and autonomous navigation (Hyyppä et al., 2021; Polewski et al., 2019; Dai et al., 2019). The quality of these maps directly determines their practical usefulness, while the cost and efficiency of their production remain critical for practical deployment. This creates a growing demand for fast, accurate, and cost-effective mapping systems that can operate reliably across diverse and challenging environments (Kukko et al., 2012; Vezeteu et al., 2025b).

Airborne laser scanning (ALS) has long been a standard solution for large-scale mapping. Mounted on aircraft or drones, ALS benefits from unobstructed Global Navigation Satellite System and Inertial Navigation System (GNSS-INS) signals, which allow for reliable georeferencing across wide areas (Wehr and Lohr, 1999; Baltsavias, 1999; Matikainen et al., 2019). However, its top-down field of view (FOV) often causes occlusions in dense environments such as forests, where tree stems and understory vegetation are shadowed by the forest canopy. Mobile laser scanning (MLS), deployed on cars, backpacks, or trolleys, provides complementary ground-level coverage and captures details that ALS misses. Yet, MLS is prone to drift: GNSS signals are frequently blocked by

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[1] https://github.com/eugeniu1994/MAP_LIO.git

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vegetation or infrastructure, and Light Detection and Ranging (LiDAR) odometry accumulates error over time (Puente et al., 2011; Kukko et al., 2012; Liang et al., 2014; Cheng et al., 2015; Kaartinen et al., 2015; Sigrist et al., 1999; Feng et al., 2021). As a result, MLS data cannot be directly integrated with ALS without additional processing.

Most existing MLS–ALS fusion approaches rely on feature-based alignment, where MLS and ALS maps are processed separately and registered using features such as roads, buildings, or trees. For instance, Cheng et al. (2015) aligned airborne and mobile LiDAR using road networks and building contours, while Hyyppä et al. (2021) employed tree stem curves for coarse registration. Lee et al. (2022) integrated MLS and ALS to estimate individual trees by fitting Digital Terrain Models (DTM) and stem curves, though their method required manual initialisation. Similarly, Muhojoki et al. (2024a) assessed MLS accuracy under forest canopies against ALS and total station data, showing that high-density ALS can achieve comparable accuracy to traditional surveying. While these methods improve map completeness (Hyyppä et al., 2020, 2021; Muhojoki et al., 2024a; Lee et al., 2022), they depend strongly on the quality and availability of features, often require manual intervention, and are not well-suited for real-time applications. In our previous work (Vezeteu et al., 2025b), we presented an online, loosely coupled MLS–ALS fusion framework in which ALS maps were used as priors for MLS registration, improving both map completeness and trajectory accuracy. In that architecture, the GNSS, MLS, and ALS modules first produced independent pose estimates, which were subsequently fused. While this design simplifies system integration, it also risks propagating individual sensor errors and failed to account for inter-sensor correlations. Furthermore, separate tunable gain parameters should be assigned to the measurements from MLS, ALS, and GNSS.

Sensor configuration also plays a critical role. On MLS platforms, multiple LiDAR sensors can be deployed to balance different strengths and weaknesses. For example, modern 3D LiDARs (e.g., Hesai, Velodyne, Ouster) provide full 6 degrees of freedom (DoF) motion observability but typically produce sparse and less accurate measurements. In contrast, 2D LiDARs (e.g., Riegl VUX) yield highly accurate and dense measurements but cannot capture full 3D motion. Their complementary nature makes direct fusion appealing, yet current methods either treat them separately or rely on external high-quality trajectories for integration. Several approaches address the limitations of 2D LiDAR. Mounting these sensors on rotating platforms enables 3D coverage (Kang et al., 2018; Schadler et al., 2014; Bosse and Zlot, 2009), but introduces motion artefacts and synchronisation issues. Alternatively, 2D scans can be projected into 3D using a high-accuracy 6 DoF trajectory from GNSS-INS (Xiaojun Li et al., 2024; Lin and Chang, 2012; Chen et al., 2017; Roca et al., 2016), though map quality then depends on trajectory accuracy and sensor calibration. In forest mapping, ALS-derived tree positions have been used to refine MLS with 2D sensors (Muhojoki et al., 2024a; Kukko et al., 2017), but this requires trunk-perpendicular scanning to observe tree arcs. In our previous work (Vezeteu et al., 2025a), we proposed a loosely coupled method for direct 3D mapping with a 2D sensor, however, it required an external $SE(3)$ initial trajectory guess and reference MLS/ALS maps, and demonstrated that 2D measurements alone could not correct noise in all DoF.

This study proposes an online, tightly coupled multi-sensor fusion framework that integrates IMU, 2D LiDAR, 3D LiDAR, GNSS-INS derived $SE(3)$, and prior ALS maps within a unified estimation platform. The system achieves robust and accurate mapping through three core components. First, we tightly fuse 2D and 3D LiDAR at the point level by directly combining their raw range measurements into a unified 3D point cloud. This strategy leverages the high angular resolution and accuracy of 2D LiDAR together with the full spatial observability of 3D LiDAR, effectively expanding the FOV and resolving the missing DoF limitations of 2D sensors. To the best of our knowledge, existing multi-LiDAR fusion methods have primarily focused on 3D–3D configurations. This work explores a registration-based fusion of 2D and 3D LiDAR in an online, tightly coupled framework. Second, we introduce a tightly coupled multi-temporal MLS–ALS fusion strategy in which prior ALS maps are incorporated in a robust, data-adaptive manner. The ALS prior mitigates long-term drift in MLS, particularly in GNSS-degraded environments, while also improving overall map completeness. The final map can be derived either from the jointly optimised point clouds or from the individual MLS or ALS. Importantly, MLS and ALS maps expressed in different reference frames are automatically initialised without manual intervention and without requiring point cloud segmentation. Finally, we develop a robust single-state maximum a posteriori (MAP) solution that jointly optimises IMU, LiDAR, $SE(3)$, and prior map constraints. The formulation employs robust, data-adaptive geometric uncertainty weighting based on local planarity estimated with Principal Component Analysis (PCA), that reduces reliance on manually tuned constant weighting parameters. The estimator fuses heterogeneous sensor information by accumulating their corresponding Hessian and gradient contributions in a unified update step. The framework is modular: the core LiDAR-inertial backbone requires only a 3D LiDAR and an IMU. A 2D LiDAR can be added as an auxiliary source of geometric constraints, while global drift can be corrected using either GNSS(-INS) or ALS prior maps. At its core, the novelty lies in casting heterogeneous multi-sensor fusion as a single-state MAP filtering problem, where all modalities contribute through a unified, data-adaptive update. An extensive ablation study further validates the contribution of each proposed component.

The main contributions of this study are:

- **Tightly coupled 2D-3D LiDAR fusion**: A novel method that directly merges complementary 2D and 3D LiDAR measurements into a unified point cloud with an expanded FOV, resulting in more stable registration, improved mapping accuracy, and natural scalability to multi-LiDAR setups.
- **Robust, tightly coupled MLS–ALS fusion**: An automatic fusion framework that integrates MLS observations with prior ALS maps using robust, data-adaptive geometric uncertainty weighting based on local planarity. Each measurement source contributes according to its estimated uncertainty. This strategy both limits global drift in MLS trajectories and increases map completeness by combining the final MLS and ALS point clouds.
- **Unified multi-sensor MAP formulation**: A tightly coupled state estimation framework that jointly fuses IMU, 3D LiDAR, 2D LiDAR, GNSS-INS derived $SE(3)$, and prior ALS constraints in real time. The proposed robust single-state MAP formulation reduces the sensor fusion problem to the accumulation of sensor-specific Hessians and gradients, weighted using robust, data-adaptive geometric uncertainty. A self-balancing mechanism normalises the influence of different sensor modalities, reducing the need for manual parameter tuning.
- **Large-scale experimental validation**: Extensive experiments over more than 5 km of mixed environment with predominant forested areas and sparse man-made structures demonstrate high mapping accuracy and computational efficiency. Ablation studies further quantify the individual impact of each proposed component.

2. Related work

Scan-based mapping relies on two key processes: registration and georeferencing. Registration aligns multiple scans into a common frame, enabling LiDAR odometry (LO), while georeferencing assigns global coordinates, typically using GNSS-INS. MLS and ALS platforms combine both, by fusing LiDAR, GNSS, and IMU, aiming to achieve precise localisation and mapping. This section reviews MLS registration methods, sensor fusion strategies, and MLS–ALS integration approaches.

2.1. MLS mapping

2.1.1. LiDAR Odometry (LO)

While feature-based methods can be more robust, they are generally slower and computationally more expensive than direct nearest-neighbour approaches. In addition, specialised descriptors have been proposed for certain environments, for example, tree-stem features in forests (Hyyppä et al., 2021, 2022), and 3D road networks and building countours in cities (Cheng et al., 2015).

The foundation of MLS registration is scan alignment, most commonly achieved using the Iterative Closest Point (ICP) algorithm and its variants (Zhang, 2014; Vizzo et al., 2023). ICP formulates registration as an optimisation problem by minimising distances between corresponding points, typically obtained via nearest-neighbour search using spatial indexing structures such as k-d trees (Cai et al., 2021) or hash maps (Vizzo et al., 2023).

Different formulations arise depending on the cost function. Point-to-point ICP (Zhang, 2014; Vizzo et al., 2023) directly minimises Euclidean distances, while point-to-plane methods (Behley and Stachniss, 2018; Chen et al., 2019) improve convergence on structured surfaces. Generalised ICP (Segal et al., 2009) extends this concept by modelling local neighbourhoods as distributions, resulting in a point-to-distribution formulation that improves robustness in noisy or unevenly sampled regions. However, it requires covariances for both clouds. While target covariances can be reliably computed from the map, source covariances estimated from the scan can be unreliable due to sensor sparsity (e.g., inner rings). Building on Generalised ICP ideas, LiDAR Odometry and Mapping (LOAM) frameworks (Zhang and Singh, 2014; Shan and Englot, 2018; Wang et al., 2020; Zhou et al., 2021) extend the pairwise point correspondences with edge and planar geometric primitives, based on the local curvature of the scan. By minimising point-to-line and point-to-plane cost functions, these systems improve stability in structured environments and form the basis of many real-time LiDAR odometry systems.

Beyond simple geometric primitives, more complex feature descriptors have also been proposed. 3D histogram-based descriptors (Scherer, 2010; Rusu et al., 2009) enable point association based on feature similarity rather than purely geometric proximity. Additionally, specialised descriptors have been designed for specific environments, such as tree-stem features in forested areas (Hyyppä et al., 2021, 2022), and road networks or building contours in urban scenes (Cheng et al., 2015). While these feature-based approaches can offer increased robustness, they are generally more computationally expensive and slower than direct nearest-neighbour methods. The optimisation step in all these registration algorithms typically minimises the residual cost using Gauss–Newton (Wang, 2012) or Levenberg–Marquardt method (Ranganathan, 2004). Registration can be performed as scan-to-scan, scan-to-map, or in hybrid forms (Zhang and Singh, 2014).

2.1.2. LiDAR-Inertial Odometry (LIO)

A key limitation of LiDAR-only odometry is motion distortion and sensitivity to poor initialisation. To address these limitations, LiDAR is often fused with IMU measurements in LiDAR-Inertial Odometry (LIO). IMU preintegration helps compensate for LiDAR motion distortion and provides initial pose estimates, while LiDAR measurements correct IMU biases and gravity estimation. Two main paradigms exist. Filtering-based methods, such as FAST-LIO and FAST-LIO2 (Xu and Zhang, 2020; Xu et al., 2022), employ an Iterated Extended Kalman Filter (IEKF) on manifolds (He et al., 2021a) for efficient state estimation, directly minimising point-to-plane residuals during the update step. Variants such as Point-LIO (He et al., 2023), built on top of Fast-LIO2, extend the framework by incorporating IMU saturation handling within the observation model. Other approaches, such as Direct LiDAR-Inertial Odometry (DLIO) (Chen et al., 2022) and Akai and Nakao (2025), combine ICP-based registration with filtering. In these methods, IMU preintegration is used for motion prediction, while the pose estimated via Generalised-ICP is used to update the system state. Pose-graph formulations have also been widely adopted (Shan et al., 2020; Kaess et al., 2011; Okawara et al., 2025; Ruyi Li et al., 2024). Methods such as LIO-SAM (Shan et al., 2020), formulate the problem as a factor graph and jointly optimise poses and IMU biases using smoothing techniques like iSAM2 (Kaess et al., 2011).

While effective, both filtering-based and pose-graph approaches rely on carefully tuned parameters. In particular, they require prior knowledge of the measurement noise covariance, which is typically manually tuned and kept fixed in practice. In contrast, the proposed method does not rely on manually specified covariance parameters, as the sensor contributions are weighted in a data-adaptive manner.

Another key factor affecting LIO performance is the limited FOV of LiDAR sensors. To improve observability, recent works aim to increase the effective FOV through active sensing or multi-sensor configurations. AEOS (Li et al., 2026) proposes adaptive LiDAR scanning for Unmanned Aerial Vehicle (UAV)-based LIO, where the sensing pattern is actively controlled to improve observability in complex scenes. By combining model predictive control with reinforcement learning, it balances exploitation and exploration, improving odometry accuracy and map completeness. Approaches such as M-LOAM (Jiao et al., 2021) extend LOAM (Zhang and Singh, 2014) to support multiple LiDARs with real-time extrinsic calibration, while MLIOM-AB (Song et al., 2024) integrates multiple 3D LiDARs with an IMU using IEKF and performs point-level fusion of the data. Similarly, M-LIO (Das et al., 2023) adopts a factor graph formulation to fuse multiple LiDARs and IMUs, improving robustness under sensor degradation or dropout.

2.1.3. 3D mapping with 2D LiDAR

While most registration methods are designed for 3D LiDAR, 2D LiDARs remain relevant for 3D mapping, due to their higher measurement density and range accuracy. However, because the mobile 2D LiDARs lack one DoF, reconstructing 3D environments with them alone is challenging.

Existing approaches for 3D reconstruction using 2D LiDAR can be broadly categorised into trajectory and mechanical-based methods. Trajectory-based methods rely on external pose estimates, commonly from GNSS-INS or MLS systems, to project 2D scans into 3D. In this context, 2D LiDARs are often integrated into MLS platforms, leveraging the MLS-provided $SE(3)$ trajectory for mapping (Xiaojun Li et al., 2024; Lin and Chang, 2012; Roca et al., 2016). While straightforward, their accuracy is strongly dependent on the quality of the trajectory and tends to degrade in GNSS-denied or degraded environments (Muhojoki et al., 2024a; Kukko et al., 2017). Our previous study (Vezeteu et al., 2025a) addressed noisy initial trajectory estimates and performed direct 3D mapping using a 2D LiDAR with ALS/MLS reference maps. The 2D scans required an initial $SE(3)$ trajectory guess for refinement, and could not correct noise in all 6 DoF.

Mechanical approaches, on the other hand, physically rotate the sensor to emulate a 3D LiDAR. For example, Sakai et al. (2017) employ a rotating 2D LiDAR to generate 3D scans, followed by Normal Distributions Transform (NDT) registration and pose graph optimisation. Similarly, Kang et al. (2018) and Bosse and Zlot (2009) mount a 2D LiDAR on rotating platforms (typically operating at 0.5–1 Hz) to enable continuous 3D scan acquisition and matching. While such systems allow the use of standard LiDAR odometry and mapping (LO/LIO) pipelines, they are inherently sensitive to mechanical inaccuracies, synchronisation issues, and motion distortion. Moreover, the low rotation frequency (0.5–1 Hz) imposes strict limitations on platform speed, making these approaches impractical for high-speed applications such as automotive scenarios.

A different approach is to combine 2D LiDAR with additional sensors for 3D mapping. For instance, Tran et al. (2021) perform 2D SLAM using a 2D LiDAR for planar pose estimation, while a 3D ultrasonic sensor is used solely for 3D mapping by projecting measurements using the estimated trajectory, without contributing to the registration

process. Unlike existing approaches that do not fully exploit 2D LiDAR measurements for complete 6 DoF motion estimation, we tightly integrate 2D and 3D LiDAR within a unified framework, enabling the 2D LiDAR to directly contribute to pose estimation rather than serving solely as a mapping sensor. Notably, while multi-LiDAR fusion has been explored, existing methods primarily focus on 3D–3D LiDAR configurations, whereas our work targets the fusion of complementary 2D and 3D LiDAR sensors.

2.1.4. LIO with GNSS integration

Although LiDAR-inertial fusion significantly improves over LiDAR-only methods, it still suffers from long-term drift. These systems usually perform well in short sequences but accumulate errors over time. Drift can be mitigated using Simultaneous Localisation and Mapping (SLAM) (Faitli et al., 2023) techniques with loop closure. However, loop closure is not always available in every scenario and increases computational overhead while introducing risks such as false loop closure correspondence. A more robust alternative is to integrate GNSS, which provides absolute position measurements that constrain drift. Both filtering and graph-based methods have successfully incorporated GNSS into LiDAR-inertial systems (Shan et al., 2020; Vezeteu et al., 2025b; Demir and Fujimura, 2019; Yang et al., 2018). Consequently, LiDAR, IMU, and GNSS are the primary sensors installed on MLS and ALS platforms. However, GNSS signals are often degraded in urban canyons and forested environments due to occlusions and multipath effects (Delamer et al., 2021; Kaartinen et al., 2015; Naesset and Jonmeister, 2002; Feng et al., 2021). As a result, while GNSS integration reduces drift, it does not fully eliminate localisation errors in these challenging environments.

2.2. MLS-ALS fusion

To address the GNSS challenges, MLS data have been used in combination with ALS, due to the complementary nature of the data. Since aerial platforms experience minimal GNSS occlusion, Post-Processed Kinematic (PPK) GNSS-INS trajectories from ALS are highly reliable. While MLS provides high-resolution, close-range measurements, ALS delivers globally consistent coverage with reliable GNSS positioning.

A common strategy for MLS–ALS registration is coarse-to-fine alignment, where an initial rough transformation is refined through iterative optimisation. In natural environments, tree-based features are widely used as stable landmarks. For instance, Hyyppä et al. (2021) proposed a method based on detecting tree stem curves in both ALS and MLS datasets to initialise alignment, followed by refinement using iterative techniques. Similarly, Lee et al. (2022) aligned tree stem curves and fitted digital terrain models (DTMs), using ICP for fine registration, although their approach required a manual initialisation. Zhou et al. (2023) further improved localisation accuracy of backpack LiDAR systems by aligning MLS data with UAV-based ALS scans using ground patches and pre-extracted tree stem features. Additional work, such as Ghorbani et al. (2024), performs co-registration of Terrestrial Laser Scanning (TLS) and ALS data by matching individual tree locations, while Muhojoki et al. (2024a) demonstrated that high-density ALS data can achieve positioning accuracy comparable to total stations under forest canopies. These findings highlight the potential of ALS as a reliable reference in GNSS-degraded environments.

In urban environments, feature-based alignment typically relies on structural elements such as road networks and building contours. More recently, learning-based approaches such as PatchAugNet (Zou et al., 2023) address heterogeneous point cloud place recognition by learning and augmenting local patch features, improving robustness to cross-source variations in density, viewpoint, and sensor characteristics. Cheng et al. (2015) proposed a hierarchical approach that performs coarse alignment using large-scale features followed by fine registration through iterative optimisation. A similar strategy was adopted by Cheng et al. (2018), focusing on building contour features for aligning ALS and TLS data. These approaches are effective in structured scenes but depend heavily on the availability of distinct geometric features.

While effective, most existing MLS–ALS fusion methods are strongly dependent on environment-specific features, such as tree stems in forests or building structures in urban areas, which limits their general applicability. In addition, they typically rely on precomputed ALS and MLS maps and operate in an offline post-processing manner to correct drift-related artefacts, such as trajectory bending in the reconstructed point clouds. To address some of these limitations, our previous work (Vezeteu et al., 2025b,a) introduced a direct method for online MLS–ALS fusion. Instead of relying on explicit feature extraction, the approach uses ALS as a prior and performs scan-to-map alignment by minimising point-to-plane residuals within an IEKF framework, demonstrating robust performance across both forested and urban environments. However, this method still depends on manually tuned noise covariance parameters in the observation model.

In contrast, the proposed approach eliminates the need for handcrafted features and avoids manual tuning of measurement covariances. It performs direct scan-to-map alignment, tightly fusing MLS and ALS data (optionally including GNSS-INS) within a unified framework. The MLS–ALS map initialisation is also handled automatically, requiring no manual intervention even when the maps are expressed in different reference frames. The system operates online at sensor frequency and remains computationally lightweight, enabling real-time performance. By formulating the problem as a single-state maximum a posteriori (MAP) estimation with robust residual fusion, the method achieves a tightly coupled integration without requiring feature detection or segmentation.

3. Method

This study introduces a tightly coupled sensor fusion approach that integrates complementary data sources, namely IMU, 2D LiDAR, 3D LiDAR, GNSS-INS derived $SE(3)$, and prior ALS. We assume all sensors are time-synchronised and extrinsically calibrated. The overall framework is illustrated in Fig. 1. The system starts by reading measurements from both the 3D and 2D LiDAR sensors and combining them into a unified point cloud. During this stage, data from the two time-synchronised LiDARs are spatially aligned using their extrinsic calibration, allowing all points to be treated as a single point cloud measurement. This fusion increases the overall point density and expands the field of view (FOV). At the same time, it leverages the superior range accuracy of the 2D LiDAR to enhance measurement quality, while compensating for its missing degrees of freedom (DoF) through the 3D LiDAR geometry. In parallel, IMU forward propagation (described in Section 3.1) is performed to provide both the reference motion required for distortion correction and an initial pose estimate for subsequent registration. Both the point cloud fusion process and the subsequent motion compensation are detailed in Section 3.2.

Next, the measurements from the MLS, ALS, and/or GNSS-INS are combined to estimate a residual vector, which is minimised in the state update. The raw 3D measured scan is simultaneously registered to both the MLS and ALS maps using the same cost function (detailed in Section 3.4). After this, the MLS map is extended with new georeferenced measurements, whereas for the prior ALS map, the current position in the map is updated. As done in our previous work (Vezeteu et al., 2025b), the ALS map is divided into 50×50 m square patches, labelled by their centre point for fast indexing. Our method simplifies the alignment process by eliminating the need for scan classification or segmentation, directly registering to downsampled 3D data. An example of prior ALS tiles are shown in Fig. C.16. The system loads four ALS tiles (forming a 100×100 m area) as a prior. When the platform moves beyond this region, the closest four tiles are dynamically loaded into memory to update the prior ALS. The ALS map itself is georeferenced

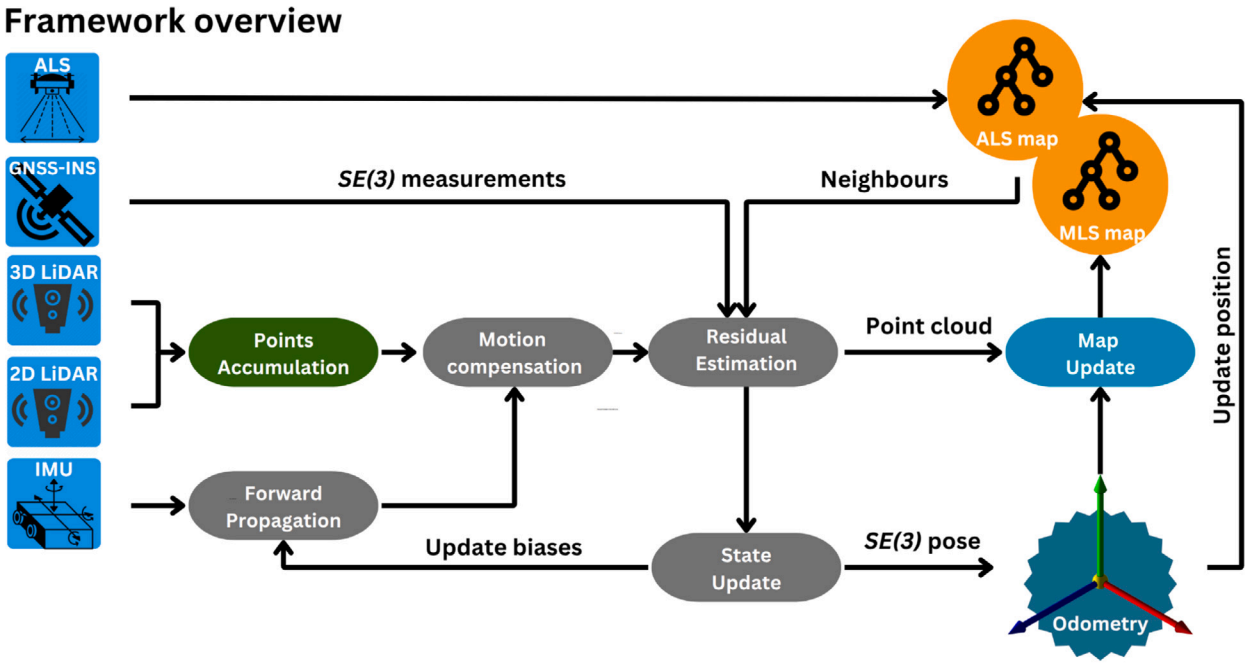


Fig. 1. The framework overview: IMU measurements are first propagated to provide an initial pose estimate and to compensate for ego-motion distortion in the accumulated points. The accumulated 3D and 2D point clouds are then synchronised and merged using the known extrinsic calibration between the sensors. Next, a nearest-neighbour search is performed on the motion-corrected scan against the MLS/ALS map to compute registration residuals. The updated state is subsequently used to correct IMU biases and to update the local MLS map with georeferenced point cloud measurements. Optionally, GNSS-INS derived $SE(3)$ measurements can be incorporated for residual computation and state updates. The residuals from MLS, ALS, and GNSS-INS are fused into a unified state estimation framework, where optimisation is performed in a tightly coupled manner.

via PPK GNSS-INS (NovAtel Inc., 2023) trajectories from aerial platforms, which benefit from minimal GNSS signal occlusion compared to ground-based MLS. ALS maps thus serve as a prior to prevent MLS global drift, while also improving map completeness through MLS–ALS fusion. Notably, while GNSS-INS can directly correct system drift in regions without GNSS signal occlusion, ALS can be considered as an indirect use of GNSS-INS since it is derived from PPK GNSS-INS trajectories.

The integration of the LiDAR-inertial system with GNSS-INS or prior ALS submaps requires initialisation, specifically the extrinsic transformation between the MLS LiDAR-inertial map ($\mathcal{M}$) and the GNSS-INS frame ($\mathcal{G}$) ${}^{\mathcal{M}}\mathbf{T}_{\mathcal{G}} = [{}^{\mathcal{M}}\mathbf{R}_{\mathcal{G}} | {}^{\mathcal{M}}\mathbf{p}_{\mathcal{G}}] \in SE(3)$. While the translational offset (${}^{\mathcal{M}}\mathbf{p}_{\mathcal{G}}$) between the GNSS antenna and the LiDAR-inertial system can be calibrated once, the orientation (${}^{\mathcal{M}}\mathbf{R}_{\mathcal{G}}$) must be re-estimated for each new session. This is because LiDAR-inertial trajectories are expressed in a local frame that varies with respect to Earth, while GNSS is provided in Earth-centred frame. As proposed in our previous work (Vezeteu et al., 2025b), we initialise this orientation by aligning a short LiDAR-inertial trajectory segment (about 30 m) with the GNSS-INS trajectory to obtain an initial transformation guess, which is later refined by co-registering overlapping parts of the MLS and prior ALS maps. Note that this approach also works when GNSS does not provide orientation. If GNSS orientation is available, the first GNSS pose can be used for an initial extrinsic estimate, which is then refined by co-registering overlapping scans. This way, the initialisation of the MLS and ALS submaps, originally expressed in different reference frames, is performed automatically using GNSS as an intermediate sensor, providing an initial alignment of the MLS state within the ALS map without manual intervention. This transformation is subsequently refined by co-registering overlapping MLS and ALS regions.

Section 3.3 introduces the residual and state update formulation and shows how the Kalman-based approach relates to the traditional least-squares solutions obtained with Gauss–Newton (Wang, 2012) and Levenberg–Marquardt algorithms (Ranganathan, 2004). Section 3.4 extends this formulation to handle GNSS-INS-derived $SE(3)$ and LiDAR point cloud measurements. Finally, Section 3.5 describes the tightly coupled state update, implemented as a maximum a posteriori problem (MAP), which jointly optimises all residuals. The notation used throughout this study is summarised in Table 1.

Table 1
Notation.

Notation	Definition
$\mathcal{M}, \mathcal{I}, \mathcal{L}, \mathcal{G}$	The MLS map, IMU, LiDAR, and GNSS frames
$\mathbf{x}, \hat{\mathbf{x}}$	The ground-truth and predicted state
$\delta\mathbf{x}$	The error between ground-truth and the estimation
$\mathbf{x}_k$	Vector $\mathbf{x}$ at time k
$\mathbf{a}_m$	Measured vector $\mathbf{a}$
${}^{\mathcal{M}}\mathbf{p}_{\mathcal{I}}$	Position of frame $\mathcal{I}$ w.r.t. frame $\mathcal{M}$ with $\mathbf{p} \in \mathbb{R}^3$
${}^{\mathcal{M}}\mathbf{R}_{\mathcal{I}}$	Rotation of frame $\mathcal{I}$ w.r.t. frame $\mathcal{M}$ with $\mathbf{R} \in SO(3)$
${}^{\mathcal{M}}\mathbf{v}_{\mathcal{I}}$	Velocity of frame $\mathcal{I}$ w.r.t. frame $\mathcal{M}$ with $\mathbf{v} \in \mathbb{R}^3$
${}^{\mathcal{I}}\mathbf{T}_{\mathcal{L}}$	Transformation of frame $\mathcal{L}$ w.r.t. frame $\mathcal{I}$ with $\mathbf{T} \in SE(3)$
$\mathcal{Q}, \mathcal{R}$	Process noise and measurement covariance.
$\mathbf{J}, \mathbf{E}$	Measurement model Jacobian and prior residual Jacobian.

3.1. State definition and propagation

We define the discrete system state in Eq. (1) as:

$$\begin{aligned} &\mathbb{M} = \mathbb{R}^3 \times SO(3) \times SO(3) \times \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^3, \\ &\mathbf{x} := \begin{bmatrix} {}^{\mathcal{M}}\mathbf{p}_{\mathcal{I}} & {}^{\mathcal{M}}\mathbf{R}_{\mathcal{I}} & {}^{\mathcal{I}}\mathbf{R}_{\mathcal{L}} & {}^{\mathcal{I}}\mathbf{p}_{\mathcal{L}} & {}^{\mathcal{M}}\mathbf{v}_{\mathcal{I}} & \mathbf{b}_{\omega} & \mathbf{b}_{a} & {}^{\mathcal{M}}\mathbf{g} \end{bmatrix}^{\top}, \\ &\mathbf{u} := \begin{bmatrix} \boldsymbol{\omega}_m & \mathbf{a}_m \end{bmatrix}^{\top}, \\ &\mathbf{w} := \begin{bmatrix} \mathbf{n}_{\omega} & \mathbf{n}_{a} & \mathbf{n}_{b_{\omega}} & \mathbf{n}_{b_{a}} \end{bmatrix}^{\top}. \end{aligned} \tag{1}$$

Here, ${}^{\mathcal{M}}\mathbf{p}_{\mathcal{I}}$, ${}^{\mathcal{M}}\mathbf{R}_{\mathcal{I}}$, and ${}^{\mathcal{M}}\mathbf{v}_{\mathcal{I}}$ denote the IMU position, orientation, and velocity in the MLS map frame $\mathcal{M}$ (with $\mathcal{M} = \mathcal{I}_0$). The IMU inputs consist of measured linear acceleration $\mathbf{a}_m$ and angular velocity $\boldsymbol{\omega}_m$, corrupted by noise terms $\mathbf{n}_a$ and $\mathbf{n}_{\omega}$. The accelerometer and gyroscope biases, $\mathbf{b}_a$ and $\mathbf{b}_{\omega}$, evolve according to zero-mean Gaussian driving noise

$\mathbf{n}_{b_a}$ and $\mathbf{n}_{b_\omega}$. The gravity vector in the map frame is represented by ${}^{\mathcal{M}}\mathbf{g}$. The dimensions of the state, input, and noise vectors are $\dim(\mathbf{x}) = 24$, $\dim(\mathbf{u}) = 6$, and $\dim(\mathbf{w}) = 12$. The state is propagated using the process model from Eq. (2):

$$\begin{aligned} &\mathbf{x}_k = \mathbf{x}_{k-1} \boxplus \left(\Delta t\ \mathbf{f}(\mathbf{x}_{k-1}, \mathbf{u}_k, \mathbf{w}_k)\right), \\ &\mathbf{x}_k \in \mathbb{M}, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathcal{Q}_k), \end{aligned} \tag{2}$$

where $\mathbf{x}_k$ denotes the state at time step k, Δt is the time increment, and $\mathcal{Q}_k$ is the process noise covariance (shown in (A.3)). The noise covariance matrix and the manifold group $\boxplus$ operator follow the definitions presented in the Appendices of our previous studies (Vezeteu et al., 2025b,a). The system dynamics $\mathbf{f}(\cdot)$ is given in Eq. (3):

$$\begin{bmatrix} {}^{\mathcal{M}}\mathbf{v}_{I_k} + \frac{\Delta t}{2}\left({}^{\mathcal{M}}\mathbf{R}_{I_k}(\mathbf{a}_{m_k} - \mathbf{b}_{a_k} - \mathbf{n}_{a_k}) - {}^{\mathcal{M}}\mathbf{g}_k\right) \\ \boldsymbol{\omega}_{m_k} - \mathbf{b}_{\boldsymbol{\omega}_k} - \mathbf{n}_{\boldsymbol{\omega}_k} \\ \mathbf{0}_{3\times 1} \\ \mathbf{0}_{3\times 1} \\ {}^{\mathcal{M}}\mathbf{R}_{I_k}(\mathbf{a}_{m_k} - \mathbf{b}_{a_k} - \mathbf{n}_{a_k}) - {}^{\mathcal{M}}\mathbf{g}_k \\ \mathbf{n}_{\mathbf{b}_{\boldsymbol{\omega}_k}} \\ \mathbf{n}_{\mathbf{b}_{\mathbf{a}_k}} \\ \mathbf{0}_{3\times 1} \end{bmatrix}. \tag{3}$$

The covariance of the state estimate $\hat{\mathbf{P}}_k$ evolves as in Eq. (4):

$$\hat{\mathbf{P}}_k = \mathbf{F}_{\mathbf{x}}\mathbf{P}_{k-1}\mathbf{F}_{\mathbf{x}}^{\top} + \mathbf{F}_{\mathbf{w}}\mathcal{Q}\mathbf{F}_{\mathbf{w}}^{\top}, \tag{4}$$

and the error-state propagation is given in Eq. (5):

$$\begin{aligned} \delta\mathbf{x}_k &= \mathbf{x}_k \boxminus \hat{\mathbf{x}}_k \\ &= \left(\mathbf{x}_{k-1} \boxplus \left(\Delta t\ \mathbf{f}(\mathbf{x}_{k-1}, \mathbf{u}_k, \mathbf{w}_k)\right)\right) \\ &\quad \boxminus \left(\hat{\mathbf{x}}_{k-1} \boxplus \left(\Delta t\ \mathbf{f}(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_k, \mathbf{0})\right)\right) \\ &\overset{\text{He et al. (2021a)}}{\cong} \mathbf{F}_{\mathbf{x}}\delta\mathbf{x}_{k-1} + \mathbf{F}_{\mathbf{w}}\mathbf{w}_k. \end{aligned} \tag{5}$$

The $\mathbf{F}_{\mathbf{x}}$ and $\mathbf{F}_{\mathbf{w}}$ are Jacobians of Eq. (5) with respect to $\delta\mathbf{x}_{k-1}$ and $\mathbf{w}_k$. The full derivation is provided in (A.1), using the approximation from He et al. (2021a).

As shown in Eq. (3), subtracting gravity from the rotated IMU acceleration requires knowledge of the initial IMU orientation. The IMU initialisation estimates both orientation and gravity, so the measured acceleration aligns with Earth's gravity. Following Vezeteu et al. (2025b), we assume 1–2 s of static data and estimate the map-frame orientation using Gram–Schmidt orthonormalisation (Leon et al., 2013).

3.2. Point cloud preprocessing

During the acquisition of a 3D scan, all IMU measurements and the 2D LiDAR points are collected in a buffer. A 3D LiDAR point cloud $\mathcal{P}_{3D}$ is acquired over the time interval $[t_0, t_e]$, where t_0 is the begin and t_e is the scan end time. Each point $\mathbf{s}_{3D}(t) = (x, y, z)^{\top}$ has an associated timestamp $t \in [t_0, t_e]$. Due to platform motion, the scan is distorted and must be motion-compensated to a common reference time t_e. All IMU measurements within $[t_0, t_e]$ are preintegrated using Eq. (2). The rigid extrinsic calibration between the 3D LiDAR and IMU is given by ${}^{I}\mathbf{T}_{\mathcal{L}} = [{}^{I}\mathbf{R}_{\mathcal{L}} | {}^{I}\mathbf{p}_{\mathcal{L}}] \in SE(3)$. The motion-compensated 3D LiDAR point expressed in the LiDAR frame at time t is computed as:

$$\mathbf{s}_{3D}^{\text{comp}} = {}^{I}\mathbf{T}_{\mathcal{L}}^{-1}\ {}^{\mathcal{M}}\mathbf{T}_{I}^{-1}(t_e)\ {}^{\mathcal{M}}\mathbf{T}_{I}(t)\ {}^{I}\mathbf{T}_{\mathcal{L}}\ \mathbf{s}_{3D} \tag{6}$$

where ${}^{\mathcal{M}}\mathbf{T}_{I}(t) = [{}^{\mathcal{M}}\mathbf{R}_{I}(t) | {}^{\mathcal{M}}\mathbf{p}_{I}(t)] \in SE(3)$ is the state in the map $\mathcal{M}$ at time t, interpolated by forward propagation from the closest discrete IMU state at time t_k (with $t_k \le t$), using a constant angular velocity and acceleration model:

$$\begin{aligned} {}^{\mathcal{M}}\mathbf{R}_{I}(t) &= {}^{\mathcal{M}}\mathbf{R}_{I}(t_k)\ \exp\left(\boldsymbol{\omega}(t_k)\,\Delta t\right), \\ {}^{\mathcal{M}}\mathbf{p}_{I}(t) &= {}^{\mathcal{M}}\mathbf{p}_{I}(t_k) + {}^{\mathcal{M}}\mathbf{v}_{I}(t_k)\,\Delta t + \frac{1}{2}\,\mathbf{a}(t_k)\,\Delta t^2, \end{aligned} \tag{7}$$

with the time interval $\Delta t = t - t_k$, $\boldsymbol{\omega}(t_k)$ and $\mathbf{a}(t_k)$ are the bias-free angular velocity and bias and gravity-free linear acceleration at t_k. $\exp(\cdot)$ is the exponential map that maps a 3D tangent vector (representing rotation axis and angle) into a finite $SO(3)$ rotation matrix (Sola, 2017).

For 2D LiDAR scans $\mathcal{P}_{2D}$, each point $\mathbf{s}_{2D}(t) = (x, y, 0)^{\top}$ is associated with a timestamp $t \in [t_0, t_e]$ and lies on the plane $z = 0$ in the 2D LiDAR frame. Using the extrinsics between the 2D and the 3D LiDAR ${}^{\mathcal{L}}\mathbf{T}_{2D} = [{}^{\mathcal{L}}\mathbf{R}_{2D} | {}^{\mathcal{L}}\mathbf{p}_{2D}] \in SE(3)$, every 2D LiDAR point $\mathbf{s}_{2D}$ is transformed to 3D LiDAR frame using:

$$\mathbf{s}_{3D} = {}^{\mathcal{L}}\mathbf{T}_{2D}\ \mathbf{s}_{2D} \tag{8}$$

and the fused 3D point cloud is:

$$\mathcal{P}_{\text{fused}} = \mathcal{P}_{3D} \cup \mathcal{P}_{2D}. \tag{9}$$

The for $\mathcal{P}_{\text{fused}}$, the Eq. (6) is applied to obtain the motion-compensated fused point cloud.

3.3. State update

This subsection introduces the measurement update step. When a sensor provides a measurement $\mathbf{z}$, the goal is to minimise the difference (residual) between the actual measurement and the predicted one:

$$\mathbf{r}(\mathbf{x}) = \mathbf{z} - \underbrace{\mathbf{h}(\mathbf{x})}_{\hat{\mathbf{z}}}, \tag{10}$$

where $\mathbf{h}(\mathbf{x})$ is the measurement function ($\hat{\mathbf{z}}$), and $\mathbf{r}(\mathbf{x})$ is the residual (innovation). Using the first order Taylor expansion around $\mathbf{x}$ and updating the state with $\delta\mathbf{x}$ gives:

$$\mathbf{r}(\mathbf{x} + \delta\mathbf{x}) \approx \mathbf{z} - \mathbf{h}(\mathbf{x} + \delta\mathbf{x}) = \underbrace{\mathbf{z} - \mathbf{h}(\mathbf{x})}_{\mathbf{r}(\mathbf{x})} - \mathbf{J}\delta\mathbf{x} = 0, \tag{11}$$

where $\mathbf{J} = \frac{\partial \mathbf{h}(\mathbf{x}+\delta\mathbf{x})}{\partial \mathbf{x}}$ is the Jacobian of $\mathbf{h}(\cdot)$. This leads to:

$$\mathbf{J}\delta\mathbf{x} = \mathbf{r}. \tag{12}$$

The solution for Eq. (12) can be found with Levenberg–Marquardt (LM) (Ranganathan, 2004):

$$\delta\mathbf{x} = \underbrace{(\mathbf{J}^{\top}\mathbf{W}\mathbf{J} + \lambda\mathbf{I})^{-1}\mathbf{J}^{\top}\mathbf{W}}_{\mathcal{A}}\mathbf{r}, \tag{13}$$

where $\mathbf{W} = \mathcal{R}^{-1}$ is the weighting matrix encoding the sensor measurement uncertainty (covariance). The parameter λ controls the LM damping. When $\lambda = 0$, it reduces to Gauss–Newton (Wang, 2012) (fast near the optimum), when λ is large, it behaves like Gradient Descent (safe far from the optimum).

The (Extended) Kalman Filter gives an alternative formulation by computing the standard Kalman gain matrix:

$$\mathbf{K} = \hat{\mathbf{P}}\mathbf{J}^{\top}\left(\underbrace{\mathbf{J}\hat{\mathbf{P}}\mathbf{J}^{\top} + \mathcal{R}}_{\mathcal{S}}\right)^{-1} \in \mathbb{R}^{\dim(\mathbf{x})\times\dim(\mathbf{z})}, \tag{14}$$

where $\mathcal{S} \in \mathbb{R}^{\dim(\mathbf{z})\times\dim(\mathbf{z})}$ is the innovation covariance and the state update is:

$$\delta\mathbf{x} = \mathbf{K}\mathbf{r}. \tag{15}$$

The standard Kalman gain formulation from Eq. (14), also known as the innovation-based form, computes the uncertainty in the measurement dimension space by inverting the $\mathcal{S}$ matrix. As shown in Xu and Zhang (2020), Vezeteu et al. (2025b), He et al. (2023), the Kalman gain can be rewritten to invert in the (smaller) state-space form instead:

$$\mathbf{K} = \underbrace{(\mathbf{J}^{\top}\mathcal{R}^{-1}\mathbf{J} + \hat{\mathbf{P}}^{-1})^{-1}}_{\in\mathbb{R}^{\dim(\mathbf{x})\times\dim(\mathbf{x})}}\mathbf{J}^{\top}\mathcal{R}^{-1} \in \mathbb{R}^{\dim(\mathbf{x})\times\dim(\mathbf{z})}. \tag{16}$$

This equivalence follows from the Woodbury identity (Riedel, 1992) (see Appendix B for derivation):

$$\hat{\mathbf{P}}\mathbf{J}^\top\left(\underbrace{\mathbf{J}\hat{\mathbf{P}}\mathbf{J}^\top+\mathcal{R}}_{\in\mathbb{R}^{\dim(\mathbf{z})\times\dim(\mathbf{z})}}\right)^{-1}=\underbrace{(\mathbf{J}^\top\mathcal{R}^{-1}\mathbf{J}+\hat{\mathbf{P}}^{-1})^{-1}}_{\in\mathbb{R}^{\dim(\mathbf{x})\times\dim(\mathbf{x})}}\mathbf{J}^\top\mathcal{R}^{-1}. \tag{17}$$

When $\dim(\mathbf{z}) < \dim(\mathbf{x})$ the innovation-based form Eq. (14) is used, otherwise the state-space form Eq. (16) is used. Note that the state-space form of the Kalman gain is analogous to $\mathcal{A}$ in Eq. (13), but with an important difference, $\lambda\mathbf{I}$ is replaced by $\hat{\mathbf{P}}^{-1}$. In LM, $\lambda\mathbf{I}$ is an unstructured isotropic prior which penalises all directions equally. It does not reflect the system dynamics and actual uncertainty. In the Kalman formulation, $\hat{\mathbf{P}}^{-1}$ is a structured prior derived from the dynamic model and past estimates. It encodes directional confidences and cross-covariances between state components of Eq. (1), making the Kalman-based update more general when a prior is available.

3.4. Application to SE(3) and point cloud measurement

In this study, we use two types of sensors for the state update, GNSS-INS and LiDAR point cloud. The GNSS-INS sensor provides the ${}^{\mathcal{G}}\mathbf{z}_k \in SE(3)$ measurement which is transformed to MLS map frame ($\mathcal{M}$) with:

$${}^{\mathcal{M}}\mathbf{z}_k = {}^{\mathcal{M}}\mathbf{T}_{\mathcal{G}}\,{}^{\mathcal{G}}\mathbf{z}_k \in SE(3), \tag{18}$$

where ${}^{\mathcal{M}}\mathbf{T}_{\mathcal{G}} = [{}^{\mathcal{M}}\mathbf{R}_{\mathcal{G}}|{}^{\mathcal{M}}\mathbf{p}_{\mathcal{G}}] \in SE(3)$ is the extrinsic transformation matrix between the GNSS frame ($\mathcal{G}$) and the MLS map frame ($\mathcal{M}$). The measurement noise is provided in the noise covariance matrix $\mathcal{R}_{\mathcal{G}} \in \mathbb{R}^{6\times6}$. Note that using $SE(3)$ is a more general case compared to only the position from GNSS. However, when GNSS does not have the orientation, the translation part can be used, as shown in our previous work (Vezeteu et al., 2025b).

The $SE(3)$ can be integrated as an absolute measurement or a relative odometry, depending on the availability of $SE(3)$ measurements or the problem. Absolute $SE(3)$ measurements constrain the pose directly in the global frame, providing a fixed reference that prevents global drift. In contrast, relative $SE(3)$ measurements do not anchor the solution globally but enforce local geometric consistency by constraining the motion between consecutive poses through relative pose measurements.

Given the absolute measured pose $\mathbf{z}_k = [{}^{\mathcal{M}}\mathbf{R}_{\mathcal{I}_k}|{}^{\mathcal{M}}\mathbf{p}_{\mathcal{I}_k}] \in SE(3)$, the residual is computed with:

$$\begin{aligned}\mathbf{r}^{\text{abs}}_{\mathcal{G}} &= {}^{\mathcal{M}}\mathbf{z}_k \boxminus \underbrace{\mathbf{h}^{\text{abs}}_{\mathcal{G}}(\mathbf{x}_k)}_{\hat{\mathbf{z}}_k}\\ &= \begin{bmatrix} {}^{\mathcal{M}}\mathbf{p}_{\mathcal{I}_k} - {}^{\mathcal{M}}\hat{\mathbf{p}}_{\mathcal{I}_k} \\ \log({}^{\mathcal{M}}\hat{\mathbf{R}}^\top_{\mathcal{I}_k}\,{}^{\mathcal{M}}\mathbf{R}_{\mathcal{I}_k})\end{bmatrix} \in \mathbb{R}^{6\times1},\end{aligned} \tag{19}$$

where $\mathbf{h}^{\text{abs}}_{\mathcal{G}}$ is the absolute $SE(3)$ measurement model with $\hat{\mathbf{z}}_k = [{}^{\mathcal{M}}\hat{\mathbf{R}}_{\mathcal{I}_k}|{}^{\mathcal{M}}\hat{\mathbf{p}}_{\mathcal{I}_k}]$ the $SE(3)$ part of the state $\mathbf{x}_k$ and $\log(\cdot)$ is the logarithm map that maps a finite $SO(3)$ rotation matrix into a 3D tangent vector (Sola, 2017). The Jacobian, derived in detail in (A.15), is given by:

$$\mathbf{J}^{\text{abs}}_{\mathcal{G}} = \frac{\partial\mathbf{h}^{\text{abs}}_{\mathcal{G}}(\hat{\mathbf{x}}_k \boxplus \delta\mathbf{x}_k)}{\partial\delta\mathbf{x}_k}\bigg|_{\delta\mathbf{x}_k=\mathbf{0}} \in \mathbb{R}^{6\times24}. \tag{20}$$

The relative $SE(3)$ measurement provides the odometry from time i to k:

$${}^{\mathcal{M}}\mathbf{z}_{ik} = {}^{\mathcal{M}}\mathbf{z}_i^{-1}\,{}^{\mathcal{M}}\mathbf{z}_k \in SE(3), \tag{21}$$

where $\mathbf{z}_{ik} = [{}^{\mathcal{M}}\mathbf{R}_{\mathcal{I}_{ik}}|{}^{\mathcal{M}}\mathbf{p}_{\mathcal{I}_{ik}}] \in SE(3)$ with $i = k-1$. The residual is computed with:

$$\begin{aligned}\mathbf{r}^{\text{rel}}_{\mathcal{G}} &= {}^{\mathcal{M}}\mathbf{z}_{ik} \boxminus \underbrace{\mathbf{h}^{\text{rel}}_{\mathcal{G}}(\mathbf{x}_k)}_{\hat{\mathbf{z}}_{ik}}\\ &= \begin{bmatrix} {}^{\mathcal{M}}\mathbf{p}_{\mathcal{I}_{ik}} - {}^{\mathcal{M}}\hat{\mathbf{p}}_{\mathcal{I}_{ik}} \\ \log({}^{\mathcal{M}}\hat{\mathbf{R}}^\top_{\mathcal{I}_{ik}}\,{}^{\mathcal{M}}\mathbf{R}_{\mathcal{I}_{ik}})\end{bmatrix} \in \mathbb{R}^{6\times1},\end{aligned} \tag{22}$$

where $\mathbf{h}^{\text{rel}}_{\mathcal{G}}$ is the relative $SE(3)$ measurement model and $\hat{\mathbf{z}}_{ik} = [{}^{\mathcal{M}}\hat{\mathbf{R}}_{\mathcal{I}_{ik}}|{}^{\mathcal{M}}\hat{\mathbf{p}}_{\mathcal{I}_{ik}}] \in SE(3)$ is the predicted relative odometry between states at time i and k:

$$\hat{\mathbf{z}}_{ik} = \begin{bmatrix} {}^{\mathcal{M}}\mathbf{R}_{\mathcal{I}_i} & {}^{\mathcal{M}}\mathbf{p}_{\mathcal{I}_i} \\ \mathbf{0}^\top & 1\end{bmatrix}^{-1}\begin{bmatrix} {}^{\mathcal{M}}\mathbf{R}_{\mathcal{I}_k} & {}^{\mathcal{M}}\mathbf{p}_{\mathcal{I}_k} \\ \mathbf{0}^\top & 1\end{bmatrix} \in SE(3). \tag{23}$$

The Jacobian for relative $SE(3)$ measurement with the full derivation provided in (A.16), is:

$$\mathbf{J}^{\text{rel}}_{\mathcal{G}} = \frac{\partial\mathbf{h}^{\text{rel}}_{\mathcal{G}}(\hat{\mathbf{x}}_k \boxplus \delta\mathbf{x}_k)}{\partial\delta\mathbf{x}_k}\bigg|_{\delta\mathbf{x}_k=\mathbf{0}} \in \mathbb{R}^{6\times24}. \tag{24}$$

The LiDAR measurement is represented as a motion-compensated voxelized point cloud $\mathcal{P}_k \in \mathbb{R}^{m\times3}$ in the LiDAR frame $\mathcal{L}$, where each point is defined by its 3D coordinates and timestamp, and m is the number of points. The point-to-plane formulation is adopted as a computationally efficient direct method that avoids explicit feature extraction or segmentation, and is less sensitive to varying point densities across different LiDAR configurations. For a given LiDAR point $\mathbf{s}_i$, with $i \in [0, m)$, the residual is:

$$\begin{aligned}\mathbf{r}_{\mathcal{L}_i} &= {}^{\mathcal{M}}\mathbf{z}^i_k - \underbrace{\mathbf{h}_{\mathcal{L}}(\mathbf{x}_k, \pi_i)}_{\hat{\mathbf{z}}^i_k}\\ &= 0 - {}^{\mathcal{M}}\boldsymbol{\nu}_i\left({}^{\mathcal{M}}\mathbf{T}_{\mathcal{I}}\,{}^{\mathcal{I}}\mathbf{T}_{\mathcal{L}}\,{}^{\mathcal{L}}\mathbf{s}_i - {}^{\mathcal{M}}\mathbf{q}_i\right) + \mathbf{n}_i \in \mathbb{R},\end{aligned} \tag{25}$$

where $\mathbf{h}_{\mathcal{L}}(\cdot)$ is the LiDAR measurement function, π_i is a plane with normal $\boldsymbol{\nu}_i \in \mathbb{R}^{3\times1}$ and point on the plane $\mathbf{q} \in \mathbb{R}^{3\times1}$, $\hat{\mathbf{z}}^i_k$ is the measurement of a plane π_i from state $\mathbf{x}_k$, ${}^{\mathcal{L}}\mathbf{s}_i$ is a source 3D point in LiDAR frame, and $\mathbf{n}_i$ is the plane measurement noise, with $\mathbf{n}_i \sim \mathcal{N}(0, \mathcal{R}_{\mathcal{L}_i})$. The transformation ${}^{\mathcal{M}}\mathbf{T}_{\mathcal{I}}$ maps the IMU body frame to the map frame (initial trajectory guess), and ${}^{\mathcal{I}}\mathbf{T}_{\mathcal{L}}$ maps the LiDAR to the IMU frame. Note that the expected measurement $\mathbf{z}_k$ is zero, since the expected value of a point-to-plane cost function is zero if a point lies precisely on the plane. The Jacobian matrix of the LiDAR point derived in (A.4), is given by:

$$\mathbf{J}_{\mathcal{L}_i} = \frac{\partial\mathbf{h}_{\mathcal{L}}(\hat{\mathbf{x}}_k \boxplus \delta\mathbf{x}_k, \pi_i)}{\partial\delta\mathbf{x}_k}\bigg|_{\delta\mathbf{x}_k=\mathbf{0}} \in \mathbb{R}^{1\times24}. \tag{26}$$

Planar measurements are obtained by querying a k-d tree built from the reference map. For each scan point, the 5 nearest neighbours are extracted and approximated with a plane using PCA. A planar measurement is accepted if its curvature $c < 0.04$, which allows sufficient geometric constraints even in weakly structured environments such as forests, where the curvature is computed from the eigenvalues of the PCA:

$$c = \frac{\lambda_0}{(\lambda_0+\lambda_1+\lambda_2)}, \quad \lambda_0 \le \lambda_1 \le \lambda_2, \tag{27}$$

where λ_i are the PCA eigenvalues. Although the tested environments could be predominantly unstructured (e.g., forest scenes), the point-to-plane formulation remains applicable because local planarity is estimated from small neighbourhoods of points. These local surface approximations provide sufficient geometric constraints even in the absence of large, well-defined planar structures. The uncertainty associated with each plane measurement explicitly accounts for two independent sources. The first source is the sensor range uncertainty, denoted by σ_{range}, which is provided by the sensor specification and is assumed to be identical for all measurements. The second source arises from the plane estimation process itself and reflects the geometric quality of the extracted plane. This plane uncertainty is characterised by the smallest PCA eigenvalue λ_0, which represents the variance along the plane normal and can be interpreted as the plane thickness. These two uncertainty components are combined to form the observation covariance of plane π_i:

$$\mathcal{R}_{\mathcal{L}_i} = \sigma^2_{\text{range}} + \sigma^2_{\text{plane},i}, \tag{28}$$

where $\sigma^2_{\text{plane},i} = \lambda_{0_i}$. Accordingly, the plane measurement likelihood is modelled as:

$$p(\mathbf{z}^i_k \mid \mathbf{x}_k) \sim \mathcal{N}\left(\mathbf{h}_{\mathcal{L}}(\mathbf{x}_k, \pi_i), \mathcal{R}_{\mathcal{L}_i}\right). \tag{29}$$

Note that this uncertainty formulation has a clear physical interpretation. While the sensor range uncertainty assigns a common noise level to all measurements, the uncertainty of the planarity adapts to the local structure of the environment. Importantly, we do not treat measurements as binary (plane/not plane), instead, measurements from well-defined, thin surfaces yield smaller eigenvalues and are therefore trusted more, whereas noisy or poorly constrained planarity (common in vegetation leaves) are naturally down-weighted. In contrast to approaches that assign a constant covariance to all plane measurements (e.g., $\mathcal{R} = 0.001$ in Xu and Zhang (2020), Xu et al. (2022), Yu et al. (2024), He et al. (2023), Vezeteu et al. (2025b)), our method treats plane measurements heterogeneously based on their intrinsic quality. This data-adaptive geometric uncertainty weighting based on local planarity improves registration robustness and removes the need for manual covariance tuning.

3.5. Tightly fusion state update

To incorporate prior information alongside the measurements, a prior residual can be defined similarly to the measurement residual $\mathbf{r}$ as $\mathbf{e} = \mathbf{x} \boxminus \hat{\mathbf{x}}$. Linearising this residual around the current estimate yields:

$$\mathbf{e}(\mathbf{x} + \boldsymbol{\delta x}) \approx (\mathbf{x} \boxplus \boldsymbol{\delta x}) \boxminus \hat{\mathbf{x}} = \mathbf{e} + \mathbf{E}\boldsymbol{\delta x}, \tag{30}$$

where the corresponding Jacobian, derived in detail in (A.10), is:

$$\mathbf{E} = \frac{\partial}{\partial \boldsymbol{\delta x}} \left[(\mathbf{x}_k^{(j)} \boxplus \boldsymbol{\delta x}) \boxminus \mathbf{x}_k^{(0)} \right]_{\boldsymbol{\delta x}=0}. \tag{31}$$

Using the prior residual $\mathbf{e}$ and the measurement residual $\mathbf{r}$, the state update is equivalent to computing a solution for the maximum a posteriori (MAP):

$$\mathbf{x}^* = \arg\min_{\mathbf{x}} \left(\|\mathbf{e}\|^2_{\hat{\mathbf{P}}^{-1}} + \|\mathbf{r}\|^2_{\mathcal{R}^{-1}} \right), \tag{32}$$

with $\mathbf{e} \sim \mathcal{N}(0, \hat{\mathbf{P}})$ and $\mathbf{r} \sim \mathcal{N}(0, \mathcal{R})$. While the IEKF-based formulations are commonly used to solve for Eq. (32) in LiDAR-inertial navigation (Xu and Zhang, 2020; Xu et al., 2022; Vezeteu et al., 2025b,a; Gao et al., 2022; Yu et al., 2024; He et al., 2021b), in this study, we employ an alternative solution that avoids explicit Kalman gain computation and instead enables direct accumulation of Hessian and gradient contributions for efficient multi-sensor fusion. These approaches are mathematically equivalent to solving the MAP problem in Eq. (32), but differ in how the solution is computed. Using the linearised prior (see Eq. (30)) and linearised measurement residuals (see Eq. (11)), the objective in Eq. (32) becomes:

$$\min_{\boldsymbol{\delta x}} \left(\underbrace{\|\mathbf{e} + \mathbf{E}\boldsymbol{\delta x}\|^2_{\hat{\mathbf{P}}^{-1}}}_{\text{prior}} + \underbrace{\|\mathbf{r} - \mathbf{J}\boldsymbol{\delta x}\|^2_{\mathcal{R}^{-1}}}_{\text{meas}} \right). \tag{33}$$

Applying Gauss–Newton, the contribution of the prior term is:

$$\boldsymbol{\delta x}_{prior} = (\mathbf{E}^\top \hat{\mathbf{P}}^{-1} \mathbf{E})^{-1} \mathbf{E} \hat{\mathbf{P}}^{-1} (-\mathbf{e}), \tag{34}$$

and the contribution of the measurement term is:

$$\boldsymbol{\delta x}_{meas} = (\underbrace{\mathbf{J}^\top \mathcal{R}^{-1} \mathbf{J}}_{\mathbf{H}})^{-1} \underbrace{\mathbf{J} \mathcal{R}^{-1} \mathbf{r}}_{\mathbf{b}}, \tag{35}$$

where $\mathbf{H}$ and $\mathbf{b}$ denote the Gauss–Newton measurement Hessian and gradient, respectively. Note that the minus sign in the prior term's gradient (Eq. (34)) arises because $\mathbf{E}$ is defined as the Jacobian of the residual $\mathbf{e}$ in Eq. (30), whereas in the measurement residual linearisation $\mathbf{r}$, $\mathbf{J}$ is defined as the Jacobian of the measurement function $\mathbf{h}(\cdot)$. Combining both terms yields the Gauss–Newton normal equation with prior, which corresponds to the information form of the single-state MAP estimate:

$$\underbrace{\left(\mathbf{J}^\top \mathcal{R}^{-1} \mathbf{J} + \mathbf{E}^\top \hat{\mathbf{P}}^{-1} \mathbf{E} \right)}_{\boldsymbol{\Lambda}} \boldsymbol{\delta x} = \underbrace{\left(\mathbf{J}^\top \mathcal{R}^{-1} \mathbf{r} - \mathbf{E}^\top \hat{\mathbf{P}}^{-1} \mathbf{e} \right)}_{\mathbf{g}} \tag{36}$$

with the solution given by:

$$\begin{aligned} \boldsymbol{\delta x}^{(j)} &= \boldsymbol{\Lambda}^{-1} \mathbf{g} \\ \mathbf{x}_k^{(j+1)} &= \mathbf{x}_k^{(j)} \boxplus \boldsymbol{\delta x}^{(j)}, \end{aligned} \tag{37}$$

where $\boldsymbol{\Lambda}$ and $\mathbf{g}$ denote the Gauss–Newton Hessian and gradient of the MAP objective. Convergence is reached when $|\boldsymbol{\delta x}^j| < \epsilon$ (with $\epsilon = 0.001$ in our experiments) or after a maximum of $j = 5$ iterations. After convergence, the state covariance is updated as:

$$\mathbf{P}_k = \boldsymbol{\Lambda}^{-1}. \tag{38}$$

The multi-sensor fusion of the data is done in terms of the measurement Hessian $\mathbf{H}$ and gradient $\mathbf{b}$ matrices as follows:

$$\begin{aligned} \mathbf{H} &= \frac{l}{m} \underbrace{\sum_{i=0}^{i<m} \mathbf{J}_{\mathcal{L}_i}^\top \mathbf{w}_i \mathbf{J}_{\mathcal{L}_i}}_{\mathbf{H}_{\text{MLS}}} + \frac{l}{n} \underbrace{\sum_{i=0}^{i<n} \mathbf{J}_{\mathcal{L}_i}^\top \mathbf{w}_i \mathbf{J}_{\mathcal{L}_i}}_{\mathbf{H}_{\text{ALS}}} + \frac{l}{1} \underbrace{\mathbf{J}_{\mathcal{G}}^\top \mathbf{w} \mathbf{J}_{\mathcal{G}}}_{\mathbf{H}_{SE(3)}}, \\ \mathbf{b} &= \frac{l}{m} \underbrace{\sum_{i=0}^{i<m} \mathbf{J}_{\mathcal{L}_i}^\top \mathbf{w}_i \mathbf{r}_{\mathcal{L}_i}}_{\mathbf{b}_{\text{MLS}}} + \frac{l}{n} \underbrace{\sum_{i=0}^{i<n} \mathbf{J}_{\mathcal{L}_i}^\top \mathbf{w}_i \mathbf{r}_{\mathcal{L}_i}}_{\mathbf{b}_{\text{ALS}}} + \frac{l}{1} \underbrace{\mathbf{J}_{\mathcal{G}}^\top \mathbf{w} \mathbf{r}_{\mathcal{G}}}_{\mathbf{b}_{SE(3)}}, \end{aligned} \tag{39}$$

where m and n denote the number of valid plane measurements in the MLS and prior ALS map, respectively, with $l = \max(m, n)$ as the maximum of the two. This weighting ensures that the Hessians and gradients from different sensor data are properly balanced. For example, each measurement has an associated weight $\mathbf{w}$. However, if the MLS map contains many more planes ($m \gg n$) than the ALS map, the MLS will dominate the state estimation. This biases the solution toward the MLS measurements, effectively over-weighting its information in the normal equations, which can lead to drift or instability, as confirmed in the ablation study. Similarly, the Hessian and gradient contributions from $SE(3)$ measurements are scaled by l, making the $SE(3)$ terms comparable in magnitude to those from LiDAR point cloud measurements. This self-balancing scaling mechanism can be interpreted as an information normalisation across sensing modalities. In the standard least-squares formulation, each measurement contributes additively to the information matrix, such that modalities with a larger number of measurements naturally dominate the solution. The proposed factors $\frac{l}{m}$ and $\frac{l}{n}$ compensate for this imbalance by normalising the total contribution of each modality with respect to its number of measurements. This is equivalent to scaling the measurement covariance, ensuring that each sensing modality contributes comparable overall information rather than being implicitly weighted by its measurement count. Therefore, the proposed scaling is not a heuristic tuning parameter, but a normalisation that ensures balanced information contribution across heterogeneous measurement sets. This is consistent with common multi-sensor fusion practices, where measurement covariances are (manually) adjusted to balance the influence of heterogeneous sensors, here, this effect is achieved automatically based on measurement counts.

The weight $\mathbf{w}$ of each measurement combines its covariance with a robust kernel as follows. Given the measurement residual $\mathbf{r}$ and covariance $\mathcal{R}$, we first compute the Mahalanobis distance:

$$d = \sqrt{\mathbf{r}^\top \mathcal{R}^{-1} \mathbf{r}} \in \mathbb{R}. \tag{40}$$

For plane measurements, this simplifies to $d = \frac{\mathbf{r}}{\sqrt{\mathcal{R}}}$. To reduce the influence of outliers, we apply the Huber (Huber, 1964) robust loss weighting:

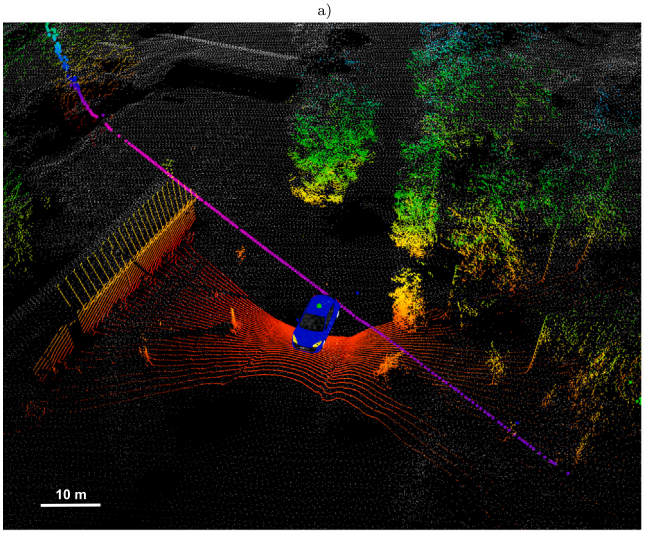
$$\rho(d) = \begin{cases} 1, & |d| \le \delta, \\ \frac{\delta}{|d|}, & |d| > \delta, \end{cases} \tag{41}$$

with threshold $\delta = 1$, corresponding to one standard deviation of a normalised Gaussian residual. The final weight is then computed as:

$$\mathbf{w} = \rho(d) \mathcal{R}^{-1} \in \mathbb{R}. \tag{42}$$



a)



b)

Fig. 2. (a) The 3D MLS system (blue frame) consists of a Hesai LiDAR sensor integrated with an IMU and a GNSS antenna, all mounted on top of the vehicle. The 2D MLS system (red frame) features a 2D Riegl VUX LiDAR, similarly paired with an IMU and a GNSS antenna. (b) Field of view (FOV) visualisation: the grey points correspond to sparse ALS data from the National Land Survey (NLS). Points from the Hesai scanner are displayed in a red-to-green colour map, while the VUX scanner points behind the vehicle are shown in purple. Different colour mappings are used for Hesai and VUX data (z-axis colouring vs. intensity) for a visual separation. A scale reference is provided in the bottom-left corner.
Source: Adapted from Vezeteu et al. (2025a).

Here, $\rho(d)$ downweights outliers, while $\mathcal{R}^{-1}$ accounts for anisotropic noise (ellipsoidal uncertainty).

The proposed multi-sensor fusion approach offers two main advantages over using a Kalman gain formulation: (i) there is no need to stack residuals and Jacobians for high-dimensional measurements, and (ii) multi-sensor fusion can be performed simply by adding the Hessian and gradient contributions from each sensor.

4. Experiments

This section describes the platform used for data collection, details the data acquisition and processing procedures, presents the configuration settings, and outlines the reference ground truth data.

4.1. Platform

The sensor system is illustrated in Fig. 2, where subfigure (a) shows the physical vehicle and (b) presents an example of the collected data along with the sensor FOV. The platform is equipped with two sensor setups mounted at the front and rear of the vehicle. The first setup, referred to as the Hesai system (Fig. 2 (a) blue frame), comprises a Hesai Pandar XT-32 LiDAR (Hesai, 2024) integrated with a CPT7 IMU (NovAtel, 2022) and a Novatel GNSS antenna (NovAtel, 2024). The Hesai system was installed to maximise point coverage in front of the vehicle. Each scan contains $\approx$ 100,000 points at 10 Hz with a measurement accuracy of ± 10 mm and precision of 5 mm (1σ). It features a $360 \times 31°$ FOV, a horizontal angular resolution of $0.18°$, and a vertical resolution of $1°$. The IMU provides inertial measurements at 100 Hz. A more detailed description of this system can be found in Vezeteu et al. (2025b).

The second system (red frame) comprises a Novatel GNSS-Inertial unit (NovAtel, 2022) paired with two Riegl VUX-1HA LiDAR scanners (RIEGL Laser Measurement Systems GmbH, 2023), although only the left VUX scanner, relative to the vehicle's travel direction, was utilised in this study. The VUX scanner operates at 250 Hz and provides high-precision point measurements with survey-grade accuracy of 5 mm, a precision of 3 mm (1σ), and an angular resolution of $0.001°$. The scanner is mounted with its laser beams oriented nearly perpendicular to the ground, enabling a detailed assessment of the road surface. Fig. 2 (b) demonstrates how the Hesai and VUX LiDAR data are merged based on temporal alignment and extrinsic calibration. The systems are considered fully calibrated. Sensor synchronisation and triggering are controlled using the GNSS clock, and the geometric transformations between the sensor frames have been accurately determined. Our previous work (Vezeteu et al., 2025a) provides a detailed description of the extrinsic calibration procedure used for the 2D and 3D LiDAR sensors.

All experiments were conducted on a laptop equipped with an Intel Core i7 processor, using a C++ implementation within the Robot Operating System (ROS) 1 framework (St-Onge and Herath, 2022).

4.2. Data collection and reference trajectory

Data collection took place on forest roads near Evo, Finland, in 2024, with the vehicle travelling at an average speed of about 40 km/h. The experiments in this study focus on a 5 km segment of the route, shown in Fig. 3.

In practice, a true ground-truth reference trajectory is rarely available. Instead, reference data are obtained from sensors with higher tested accuracy, often validated in laboratory settings and under human supervision. For this study, we have a semi-manual ground-truth reference trajectory generated by combining information from multiple sources, to reach better accuracy than PPK GNSS/INS, commonly used as a reference by other authors.

The reference trajectory leverages the high positional accuracy ALS dataset, collected in 2023 with a helicopter-mounted Riegl VUX-1HA LiDAR (RIEGL Laser Measurement Systems GmbH, 2023) and a PPK GNSS-INS solution processed with NovAtel Inertial Explorer (NovAtel, 2024; NovAtel Inc., 2023). This ALS dataset achieved an accuracy of $\approx$ 5 cm RMSE for derived tree locations (Muhojoki et al., 2024b,a; Vezeteu et al., 2025b) and served as the prior spatial reference. The vehicle trajectory was then aligned to the reference ALS using the PPK GNSS-INS solution from the sensors mounted on the car roof. This provided an initial estimate for positioning each Hesai LiDAR scan within the reference ALS map. The scans were subsequently motion-compensated using IMU data and co-registered to the reference ALS map to refine the initial PPK GNSS-INS alignment. Finally, both the registration results and the reference map were manually inspected to identify anomalies and assess overall localisation quality. Notably, this reference ALS

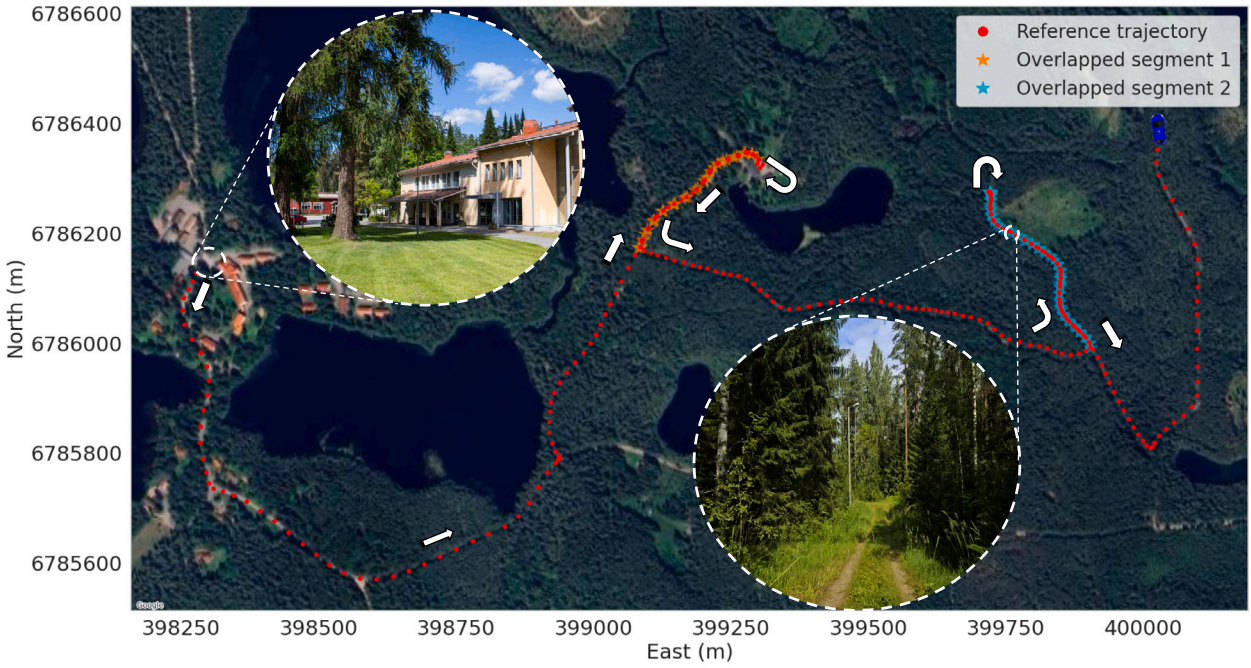


Fig. 3. Orthophoto of the Evo region in Southern Finland, where the ablation study was carried out. The red dots represent the reference ground truth positions (plotted every 40th point). The orange and cyan stars show trajectory parts with overlap. White arrows on the left side of the trajectory show the direction. The close-up white circles illustrate an example of the environment visualisation.

dataset was used solely for generating the reference trajectory and was not employed in any other experimental configurations.

Additional validation for the reference trajectory was conducted using overlapping path segments. The vehicle traversed the same narrow, single-lane road in both forward and backward directions. Consequently, the z-axis values should ideally remain consistent, as they are constrained by the ground surface (as illustrated in the close-up image Fig. 3). For each overlapping segment, the z-axis measurements from the forward pass were compared against those from the backward pass. Two such overlapping trajectory segments were identified (corresponding to the orange and cyan trajectories in Fig. 3), covering a total travelled distance of 621 m. The mean z-axis error of the reference trajectory was 0.04 m, with a standard deviation of 0.03 m.

A natural question arises: why not always rely on this approach to obtain accurate trajectory data for georeferencing point clouds? While effective, the method is both time-consuming and costly. Moreover, reference ALS datasets are not always available for every area of interest, and the process requires substantial manual effort and supervision. These limitations restrict its scalability and make it unsuitable for large-scale or long-duration datasets. This is why we developed the proposed automatic method.

4.3. Test scenarios and evaluation

This subsection describes the tested configurations and evaluation methods. Two types of ALS prior maps, illustrated in Fig. 4, were tested in the experiments to mitigate global drift. It is important to note that these ALS datasets differ from the one used to generate the reference data, which served solely to establish the reference trajectory.

The first prior map was a sparse ALS dataset (Fig. 4 a) with an average of $\approx$ 5 points/m^2, provided by the National Land Survey of Finland National Land Survey of Finland (2024), with a maximum vertical error of 10 cm and a planimetric error of up to 45 cm. We refer to this map as 'S-ALS'. This dataset is available all over Finland (most European countries already maintain ALS data, with more adopting it). The second prior map was a dense ALS point cloud (Fig. 4 b) collected by the Finnish Geospatial Research Institute (FGI) in 2024, georeferenced using a PPK GNSS-INS solution processed with RiPROCESS (RIEGL Laser Measurement Systems GmbH, 2023), with an approximate point density of $\approx$ 5000 points/m^2. This dataset reports an absolute accuracy of up to 20 cm and an RMSE of $\approx$ 5 cm, and it is referred to as 'D-ALS'. The availability of this dataset is restricted by the FGI and is not publicly accessible.

The method proposed in this study enables flexible multi-sensor fusion. To systematically evaluate its performance, we start with a basic LiDAR–inertial system and progressively incorporate additional sensor measurements, including a 2D VUX LiDAR, a prior ALS map, and $SE(3)$ measurements. The tested configurations are summarised as follows:

- **GNSS-INS**: An $SE(3)$ trajectory obtained from the PPK GNSS-INS solution using Inertial Explorer (NovAtel Inc., 2023). This configuration serves as a baseline for georeferencing the point cloud and assessing the performance of PPK GNSS-INS on forest roads.
- **LI**: A tightly coupled LiDAR-inertial system combining a 3D Hesai LiDAR with an IMU using the method proposed in this study. This configuration demonstrates that, while LiDAR-inertial fusion provides accurate local motion estimation, it exhibits drift over longer trajectories.
- **LI-VUX**: An extension of the 'LI' configuration that additionally integrates measurements from the 2D VUX LiDAR into a single point cloud, as described in Section 3.2. This setup evaluates whether the increased point density and expanded FOV improve overall system performance.
- **LI-VUX + $SE(3)$**: The 'LI-VUX' system augmented with absolute $SE(3)$ measurements from the PPK GNSS-INS solution. This configuration investigates how absolute $SE(3)$ measurements can correct long-term drift in the LiDAR-inertial system.

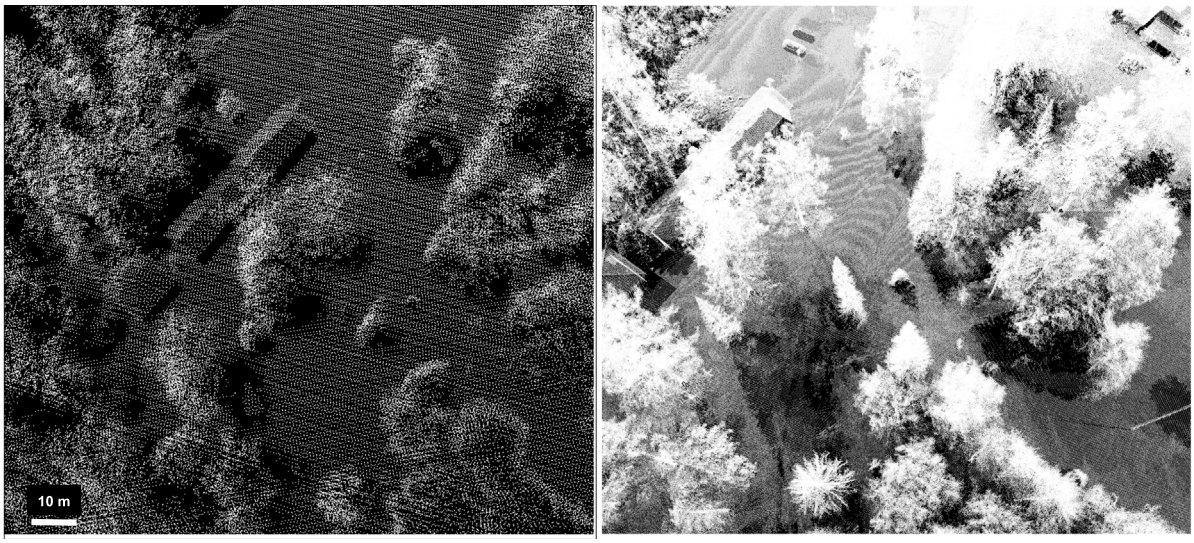


Fig. 4. Top-down comparison between sparse and dense ALS data at the start of the measurement campaign in Evo, Southern Finland (61.19647° N, 25.10744° E). (a) Sparse ALS (S-ALS) point cloud provided by the National Land Survey of Finland National Land Survey of Finland (2024), with an average density of 5 points/m^2. (b) Dense ALS (D-ALS) point cloud collected by the Finnish Geospatial Research Institute (FGI) in 2024, with a density of ≈ 5000 points/m^2. The scale is shown in the bottom-left corner.

- **LI-VUX + S-ALS**: The configuration proposed in this study, in which 3D Hesai and 2D VUX LiDAR measurements are tightly fused with IMU data and sparse ALS priors.
- **LI-VUX + S-ALS + $SE(3)$**: The 'LI-VUX + S-ALS' system further augmented with GNSS-INS-derived $SE(3)$ relative measurements. Absolute $SE(3)$ measurements are not used here, as the prior ALS map already provides absolute positioning that may not align perfectly with GNSS-INS, instead, $SE(3)$ is incorporated as relative odometry constraints.
- **LI-VUX + D-ALS**: Similar to 'LI-VUX + S-ALS', but using a dense ALS map as a prior. This experimental configuration assesses the impact of prior map quality on system performance.
- **LI-VUX + D-ALS + $SE(3)$**: The 'LI-VUX + D-ALS' system augmented with relative GNSS-INS-derived $SE(3)$ relative measurements.

Note that although the 'LI-VUX + D-ALS' configuration is evaluated in this experiment, it is of limited practical relevance. From a real-world perspective, it is less feasible than using a sparse ALS prior due to the high cost and limited availability of dense ALS data. Moreover, since this map is already dense, it does not significantly benefit from the additional point density provided by MLS. It is evaluated to demonstrate that the proposed system's performance improves with higher-quality prior maps.

The estimated trajectories were evaluated using Absolute Trajectory Error (ATE, m) for global drift and Relative Trajectory Error (RTE, % per 100 m) for local consistency, computed with the Evaluation of Odometry (EVO) package (Grupp, 2017). Given the reference and estimated poses $\mathbf{T}_i^{ref}, \mathbf{T}_i^{est} \in SE(3)$, the ATE is computed with:

$$\text{ATE} = \|(\text{trans}((\mathbf{T}_i^{ref})^{-1}\mathbf{T}_i^{est})\|_2, \tag{43}$$

where trans$(\cdot)$ extracts the 3D translation vector from $SE(3)$ matrix. The RTE are computed for pose pairs (i, j) separated by 100 m travelled distance:

$$\Delta\mathbf{T}_{ij}^{ref} = (\mathbf{T}_i^{ref})^{-1}\mathbf{T}_j^{ref}, \quad \Delta\mathbf{T}_{ij}^{est} = (\mathbf{T}_i^{est})^{-1}\mathbf{T}_j^{est}, \tag{44}$$

and the error is computed from the full transformation as:

$$\text{RTE} = \|(\text{trans}((\Delta\mathbf{T}_{ij}^{ref})^{-1}\Delta\mathbf{T}_{ij}^{est})\|_2. \tag{45}$$

All estimated trajectories were aligned with the reference trajectory to compute the ATE. For each setup, we report mean, median, root mean square error (RMSE), and standard deviation, and assess z-axis consistency by comparing forward and backward overlaps.

4.4. Ablation study

In this experiment, we evaluated the proposed LiDAR-inertial ('LI') configuration and compared it against two state-of-the-art methods, Fast Direct LiDAR-inertial Odometry (FAST-LIO2) (Xu et al., 2022) and Direct LiDAR-Inertial Odometry (DLIO) (Chen et al., 2022), using their publicly available implementations.[2,3] FAST-LIO2 is a tightly coupled filtering approach that performs scan-to-map registration by fusing IMU propagation with LiDAR planar constraints using the IEKF formulation. The innovation term is based on point-to-plane residuals, using a constant measurement covariance $\mathcal{R} = 0.001$ m^2 for all planes. DLIO follows a scan-to-map approach and uses Generalised ICP (G-ICP) (Segal et al., 2009) estimation in the measurement model. It applies separate tunable gains to each state component: $K_p = 4.5$ (position), $K_v = 11.25$ (velocity), $K_q = 4.0$ (orientation), $K_{ab} = 2.25$ (accelerometer bias), and $K_{gb} = 1.0$ (gyroscope bias), as specified in the source code. According to their public implementations, FAST-LIO2 downsamples each scan using a voxel grid with a 0.5 m leaf size prior to registration, while DLIO uses a 0.25 m leaf size. In all our experiments, LiDAR point clouds were downsampled using a voxel grid with a leaf size of 0.5 m.

For the proposed 'LI' configuration, we evaluate the impact of different measurement weighting strategies:

- **Our (**):** This setting is equivalent to the covariance model used in Fast-LIO2, which relies solely on sensor range uncertainty, with range noise $\sigma^2_{\text{range}} = 0.001$ m^2, resulting in a constant covariance $\mathcal{R} = \sigma^2_{\text{range}}$.

[2] https://github.com/hku-mars/FAST_LIO
[3] https://github.com/vectr-ucla/direct_lidar_inertial_odometry

Table 2
Absolute and relative trajectory error statistics. **Blue** colour highlights the best results (smallest errors), and **bold** numbers show the second best results.

Statistics	ATE (m)				RTE (%)			
	Mean	Median	RMSE	Std	Mean	Median	RMSE	Std
GNSS-INS	0.415	0.382	0.46	0.197	2.509	2.159	2.888	1.431
LI	7.043	6.332	7.886	3.547	0.564	0.427	0.699	0.413
LI-VUX	6.009	5.847	6.716	2.999	0.431	0.376	0.508	0.268
LI-VUX + $SE(3)$	0.363	0.286	0.415	0.201	0.399	0.341	0.465	0.238
LI-VUX + S-ALS	0.188	0.183	0.206	0.085	0.396	0.366	0.459	0.231
LI-VUX + S-ALS + $SE(3)$	0.171	0.16	0.194	0.091	0.376	0.33	0.435	0.218
LI-VUX + D-ALS	**0.036**	**0.031**	**0.045**	**0.027**	**0.173**	**0.14**	**0.22**	**0.136**
LI-VUX + D-ALS + $SE(3)$	**0.039**	**0.034**	**0.049**	**0.028**	**0.197**	**0.164**	**0.244**	**0.145**

- **Our (*):** This configuration combines the sensor range uncertainty σ_{range} with the plane local geometric uncertainty estimation with PCA σ_{plane}, as defined in Eq. (28). The plane uncertainty σ_{plane} is estimated from local planarity that varies across individual plane measurements.
- **Our best:** This configuration further augments the combined sensor and plane uncertainties with a robust kernel, as described in Eq. (41). The final setup jointly exploits data-adaptive geometric uncertainty from Eq. (42) and robust weighting, requiring no manual parameter tuning since all parameters are inferred directly from the data.

For all experiments, only LiDAR inertial odometry was used, without any prior ALS or $SE(3)$ measurements.

While most configurations in our experiments use the combined 3D and 2D LiDAR setup (LI-VUX), the second ablation study evaluates the contribution of the 2D VUX sensor in the final system, by comparing registration accuracy with and without the additional 2D LiDAR.

Another ablation experiment investigated the effect of omitting the scaling factors described in Eq. (39) when fusing LiDAR-inertial solutions with prior ALS or absolute $SE(3)$ measurements. This demonstrates the bias of the solution toward the MLS measurement and motivates the proposed scaling technique.

Finally, we compared loosely coupled versus tightly coupled MLS–ALS integration. In this study, we proposed a tightly coupled approach, where MLS and prior ALS measurements are fused simultaneously. In contrast, our previous work (Vezeteu et al., 2025b) used a loosely coupled approach, where MLS and prior ALS data were fused iteratively using IEKF: first registering scans to MLS and then to ALS maps. For this comparison, we tested our 'LI-VUX + S-ALS' configuration against a loosely coupled version, referred to as 'LI-VUX + S-ALS (l-coupled)'. Unlike our previous work, the loosely coupled version here includes the 2D VUX LiDAR and the proposed robust data-adaptive geometric uncertainty to ensure a fair comparison between the two implementations. We report both the ATE and RTE, and additionally assess computational performance. The runtime comparison between the loosely coupled ('LI-VUX + S-ALS (l-coupled)') and the proposed tightly coupled MAP-based ('LI-VUX + S-ALS') methods was conducted by executing each configuration 10 times on the same dataset.

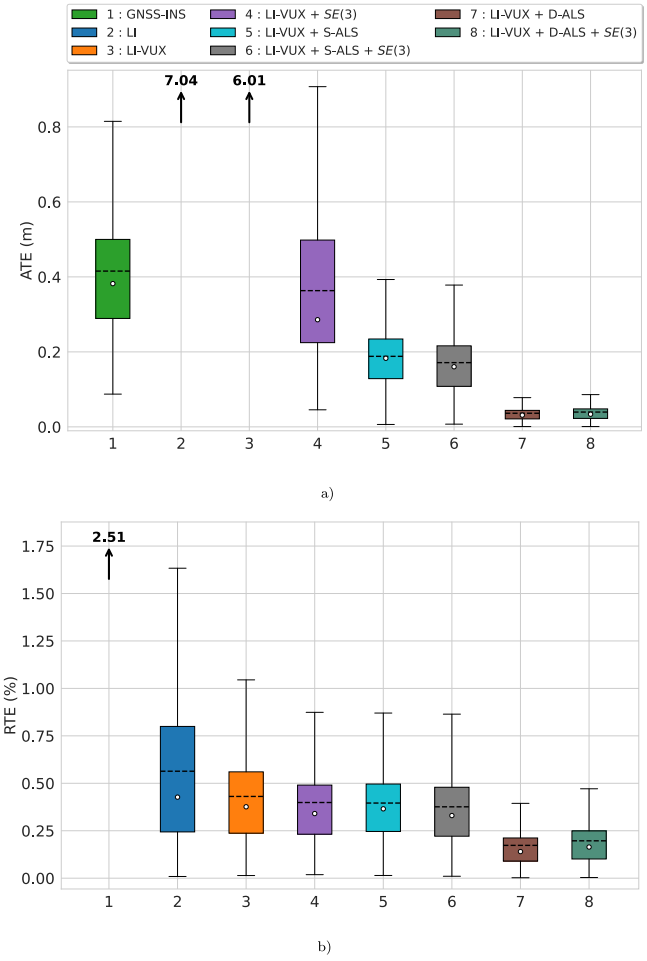


Fig. 5. Box plots showing the trajectory errors for each configuration: (a) absolute trajectory error (ATE), (b) relative trajectory error (RTE) per 100 m. The dashed black line indicates the mean value in each configuration, while the white circle shows the median. For clearer visualisation, 'LI' and 'LI-VUX' are omitted from the ATE plot, and 'GNSS-INS' from the RTE plot, the corresponding statistics are provided in Table 2.

5. Results and discussions

5.1. Quantitative analysis

The ATE and RTE statistics are summarised in Table 2. For a quick visual overview of the trends across configurations, Fig. 5 provides an intuitive representation. In the following, we discuss the results based on the mean error values.

The 'GNSS–INS' configuration provides an absolute error of 0.4 m, substantially lower than the 7.04 m observed for the LiDAR-inertial ('LI') system. However, the 'LI' configuration achieves a smaller relative trajectory error of 0.56% compared to 2.5% for 'GNSS–INS'. Integrating the Hesai and VUX scans into a unified point cloud ('LI-VUX') improved localisation accuracy compared to the Hesai-only 'LI' configuration: the absolute error decreased from 7.04 m to 6 m, and the relative error dropped from 0.56% to 0.43%. While this improvement demonstrates the benefit of increased point density and a wider FOV, the results remain insufficient for high-precision applications, indicating that additional processing is needed to further reduce drift. Overall, these findings highlight that GNSS-INS provides superior absolute

accuracy, whereas the LiDAR-inertial solution offers stronger local consistency, motivating the fusion of both approaches to combine their complementary strengths.

The 'LI-VUX + $SE(3)$' configuration fuses the LiDAR-inertial system ('LI-VUX') with absolute $SE(3)$ measurements from GNSS-INS. This integration combines the absolute positioning accuracy of GNSS-INS with the local precision of the LiDAR-inertial system, resulting in an absolute error of 0.36 m and a relative error of 0.4%.

Further improvements in absolute error were achieved by incorporating prior ALS maps. Integrating the LiDAR-inertial system with sparse ALS data ('LI-VUX + S-ALS') reduced the absolute error to 0.18 m, and the relative error remained at 0.4%. The absolute error slightly decreased when combining the LiDAR-inertial system with both the prior ALS map and $SE(3)$ measurements in the 'LI-VUX + S-ALS + $SE(3)$' configuration. Here, $SE(3)$ measurements are incorporated as relative odometry to avoid misalignment with the absolute prior ALS map, resulting in an absolute error of 0.17 m and a relative error of 0.37%.

The highest accuracy was achieved when the system was fused with a dense prior ALS map in the 'LI-VUX + D-ALS' configuration, resulting in an absolute error of 0.036 m and a relative error of 0.173%, effectively matching the reference trajectory with similar accuracy. While this demonstrates that high accuracy is attainable, it relies on the availability of a highly precise prior map. For practical applications, we rely on a sparse ALS map, which offers broad geographic coverage despite being slightly less accurate.

The further inclusion of relative $SE(3)$ measurements in the 'LI-VUX + D-ALS' configuration, evaluated as 'LI-VUX + S-ALS + $SE(3)$', did not lead to any improvement over using the dense ALS prior alone. In fact, the results became slightly worse, with ATE increasing from 0.036 m to 0.039 m and RTE from 0.173% to 0.197%. While the fusion of relative $SE(3)$ constraints with sparse ALS data (S-ALS) showed improvements in earlier configurations, this benefit was not observed when higher-quality ALS data was available. This behaviour may be attributed to overoptimistic covariances of the $SE(3)$ constraints, or to the fact that the evaluated configurations already approach the accuracy of the reference trajectory, making further improvements difficult to demonstrate given this reference.

5.1.1. *Z-axis error analysis*

Fig. 6 illustrates the z-axis errors for the overlapping trajectory segments. A modest improvement is observed when incorporating the 2D VUX scanner into the LiDAR-inertial system ('LI-VUX'), where the mean overlap z-axis error is reduced from 0.25 m to 0.22 m. Fusing absolute $SE(3)$ measurements from GNSS–INS ('LI-VUX + $SE(3)$') does not further reduce the mean z-axis error compared to GNSS-INS alone, with both configurations achieving a mean error of 0.15 m, however, the median error decreases from 0.12 m to 0.09 m.

Consistent with the trends observed in the absolute and relative trajectory errors, the best performance is achieved when prior ALS data are incorporated. In particular, the 'LI-VUX + S-ALS + $SE(3)$' configuration yields a marginal improvement of $\approx$ 1 cm, reducing the z-axis error from 0.04 m ('LI-VUX + S-ALS') to 0.03 m. The use of a dense ALS prior further improves performance, achieving a z-axis error of 0.02 m. The cumulative distribution functions (CDFs) of the z-axis errors shown in Fig. 7 further support these observations. Notably, when dense ALS data are used as prior maps, the 'LI-VUX + D-ALS' configuration even outperforms the reference trajectory for the overlapping segments according to this error metric.

In summary, the quantitative analysis shows that fusing measurements from the 3D Hesai and 2D VUX scanners into a unified point cloud improves overall system performance. However, this fusion alone does not fully eliminate global drift, which can be mitigated using GNSS-INS measurements. High accuracy with GNSS-INS is possible with PPK signals, making it an offline solution. In contrast, leveraging prior ALS maps allows the system to operate fully online and yields even

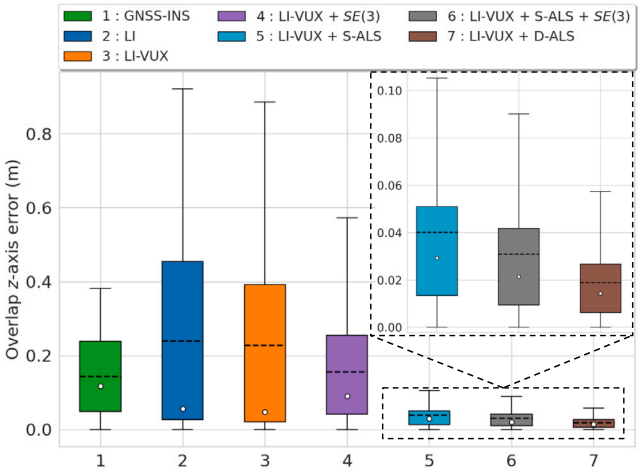


Fig. 6. Box plots showing the overlap z-axis errors for each configuration. The dashed black line indicates the mean value, while the white circle shows the median.

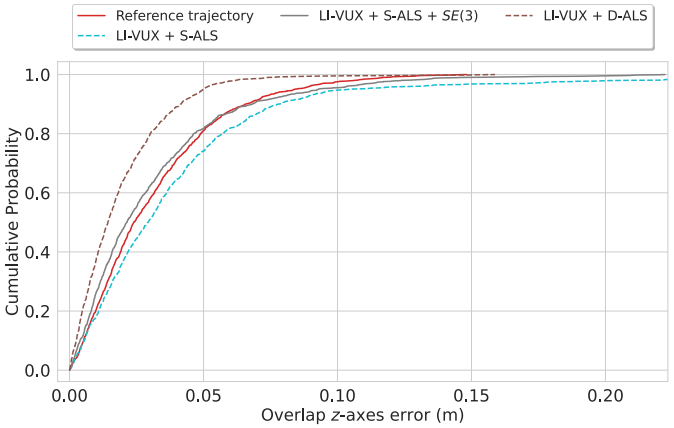


Fig. 7. The cumulative distribution function (CDF) of z-axis errors.

better results. Nevertheless, in the absence of dense ALS data, even a sparse ALS dataset, with a point density of $\approx$ 5 points/m^2, proved sufficient to prevent most trajectory drift. However, while the use of sparse ALS shows smaller absolute errors, on the relative scale, both the configuration with GNSS-INS $SE(3)$ and the one using the sparse ALS prior with relative $SE(3)$ measurements exhibit similar relative errors, as shown in Fig. 5 (b).

5.2. *Qualitative analysis*

Fig. 8 presents qualitative results from a rural area, depicting a section of the road at the beginning of the test trajectory. The comparison between maps generated using Hesai and VUX scans highlights the higher point density and finer structural detail captured by the 2D VUX scanner relative to the 3D Hesai sensor. The denser VUX measurements also enable more accurate assessments of road surface quality. These observations motivate the integration of both modalities to enhance overall mapping performance.

Similarly, Fig. 9 shows results obtained on a forest road using the 'LI-VUX + D-ALS' configuration, which produces sharp, high-quality maps even in this challenging environment. In vegetated areas, scanner density plays a crucial role, as the 2D VUX captures fine vegetation structures, such as thin branches, that are not visible in the Hesai

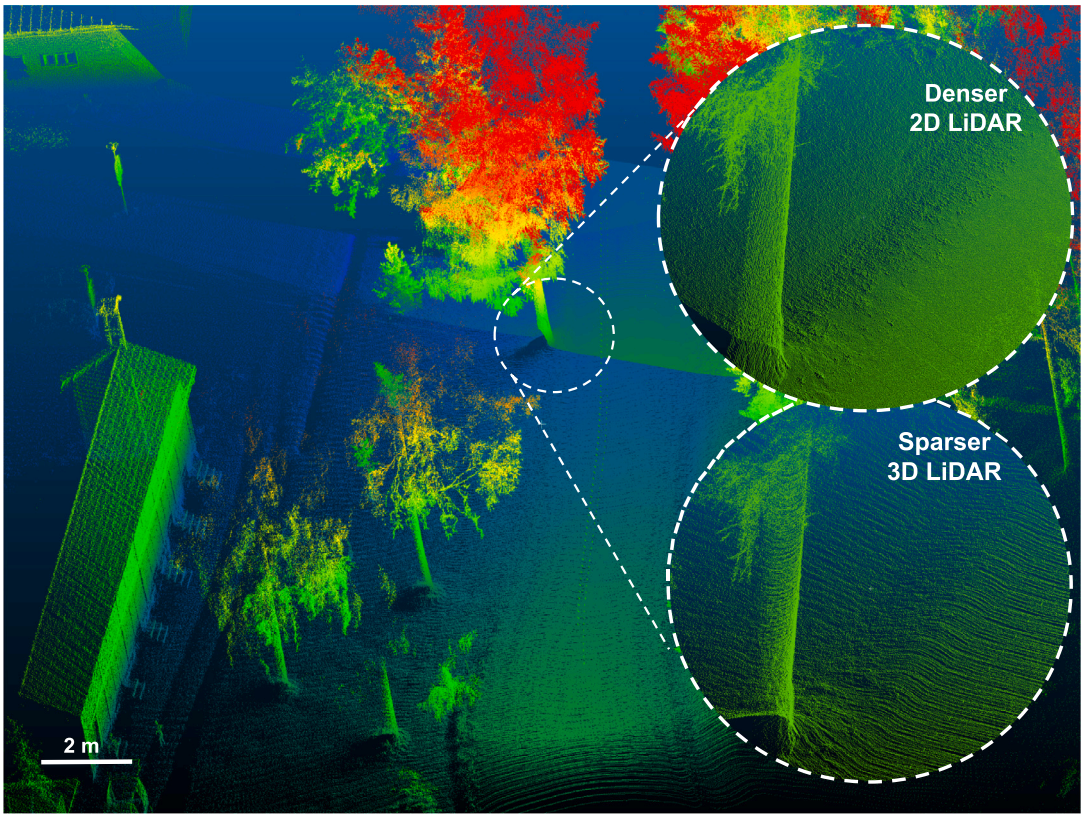


Fig. 8. Comparison of point cloud density between the 3D Hesai and 2D VUX scanners. The figure highlights how the addition of the VUX scanner increases point coverage and improves spatial detail in the generated maps. The point cloud is coloured by the z-axis. The scale is shown in the bottom-left corner.

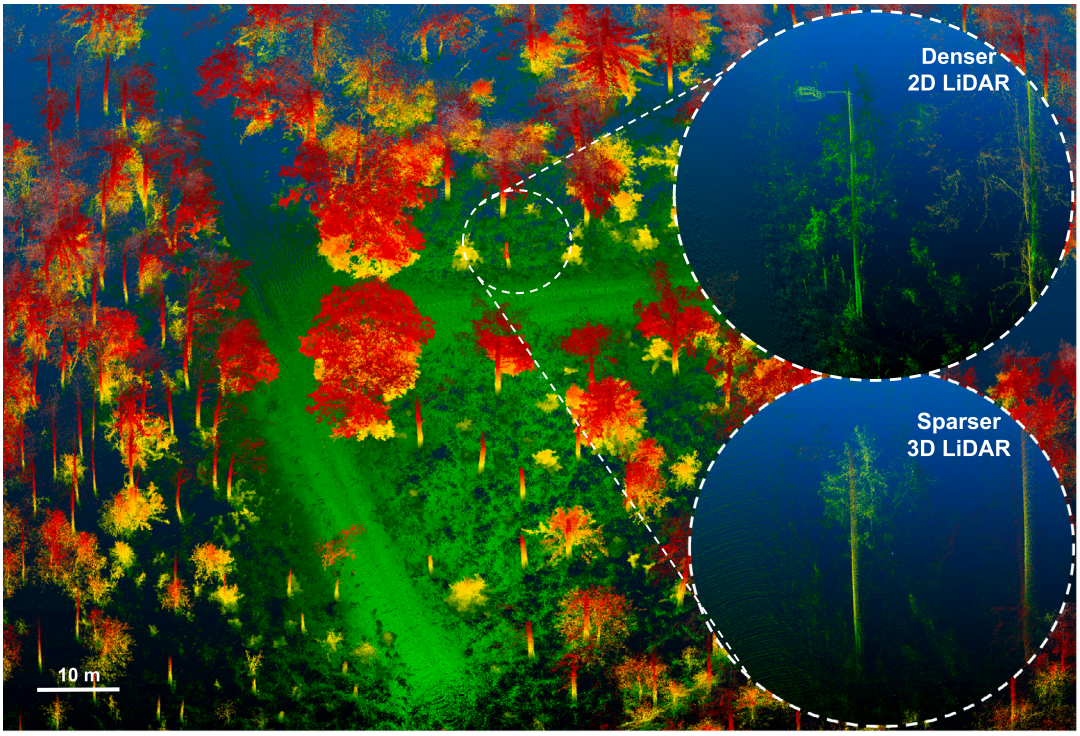


Fig. 9. Qualitative point cloud results on a forest road. The close-up windows compare the areas scanned using the 3D Hesai and 2D VUX sensors, highlighting differences in vegetation representation. The point cloud is coloured by height (z-axis), and the scale is shown in the bottom-left corner.

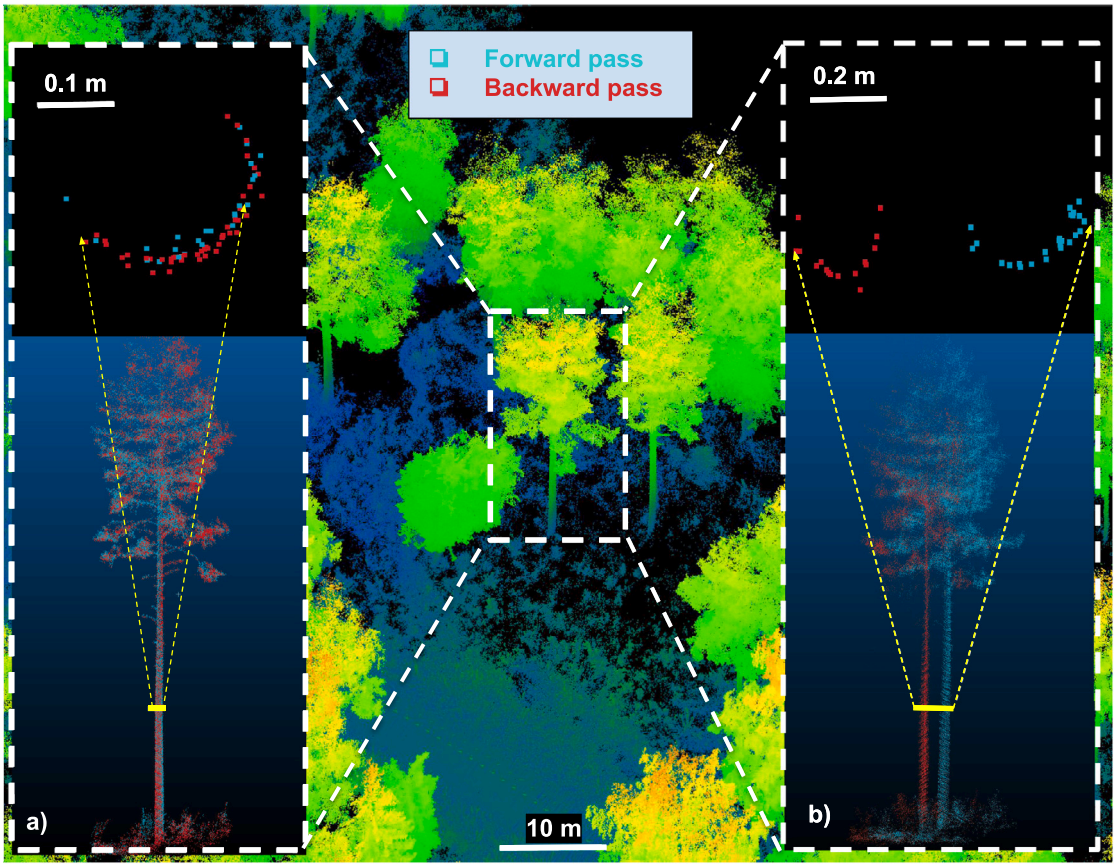


Fig. 10. Cross-sectional views of tree trunks in overlapping regions between forward and backwards passes, highlighting misalignment and demonstrating improvements. (a) Tree and corresponding cross-sectional trunk estimates obtained with our 'LI-VUX+S-ALS' method. (b) corresponding estimates obtained using the PPK 'GNSS-INS' solution. The main point cloud is coloured by height (z-axis) with the scale shown at the bottom, while in (a) and (b), points are coloured cyan and red for forward and backward passes, respectively.

scans. The resulting point cloud provides improved detail and sharpness, which is advantageous for forestry applications, including tree segmentation and classification, by supplying higher-quality input data.

Further qualitative results are presented in Fig. 10, which illustrates the overlap between point clouds acquired during the forward and backward passes. The figure compares the reconstruction obtained using our 'LI-VUX+S-ALS' approach with that generated from the PPK 'GNSS-INS' solution. A clear degradation is observed in the GNSS-INS-based reconstruction, where trajectory inconsistencies lead to duplicated tree structures. In addition, cross-sectional views of the tree trunks, extracted at a height of 1 m, are shown to provide a more detailed assessment of alignment accuracy. These cross sections further highlight the discrepancies in the GNSS-INS solution, where noticeable offsets and distortions are present. In contrast, the proposed 'LI-VUX+S-ALS' method yields well-aligned and consistent trunk geometries, demonstrating its improved robustness and accuracy in overlapping regions.

5.3. Ablation study analysis

Fig. 11 summarises the errors obtained with the state-of-the-art LiDAR-inertial odometry methods Fast-LIO2 and DLIO, and compares them with the proposed approach. The ablation study evaluates three variants of our method: 'Our(**)', which uses only the sensor range covariance as explained in the ablation study Section 4.4, 'Our(*)', which combines sensor range covariance with data-adaptive geometric uncertainty weighting, and 'Our best', which corresponds to the full proposed method combining robust kernels with sensor range and data-adaptive geometric uncertainty weighting. A clear performance trend is observed from 'Our(**)' to 'Our best'. For the 'LI' configuration, the absolute error decreases from 27 m to 7.04 m, while the relative error is reduced from 2% to 0.56%. Among the baseline methods, DLIO outperforms Fast-LIO2 in both absolute and relative errors, however, both methods yield inferior performance on the evaluated dataset when compared to the proposed 'Our best' configuration.

The role of the 2D VUX sensor in the final system is illustrated in Fig. 12. As shown in previous results, in simpler configurations, such as the LiDAR-inertial (LI) setup, the inclusion of the VUX sensor leads to noticeable improvements, decreasing the absolute trajectory error by approximately 1 m and the relative error by 0.1%. However, in the full system configurations, the impact of the VUX sensor on trajectory accuracy becomes marginal. For instance, in setups using only $SE(3)$ constraints, the inclusion of the VUX sensor yields a minor improvement of approximately 0.02 m in absolute trajectory error, with negligible changes in relative error. Similar trends are observed in configurations incorporating 'S-ALS' and 'S-ALS + $SE(3)$', where the improvement in absolute error is limited to 0.02–0.03 m and relative errors remain largely unchanged. In the final configuration, which utilises high-quality dense ALS data ('D-ALS'), no measurable difference in trajectory accuracy is observed between configurations with and without the VUX sensor.

These results indicate that as additional sensing modalities are integrated, or when high-quality prior maps are available, the contribution of the 2D VUX sensor to trajectory accuracy diminishes. Nevertheless,

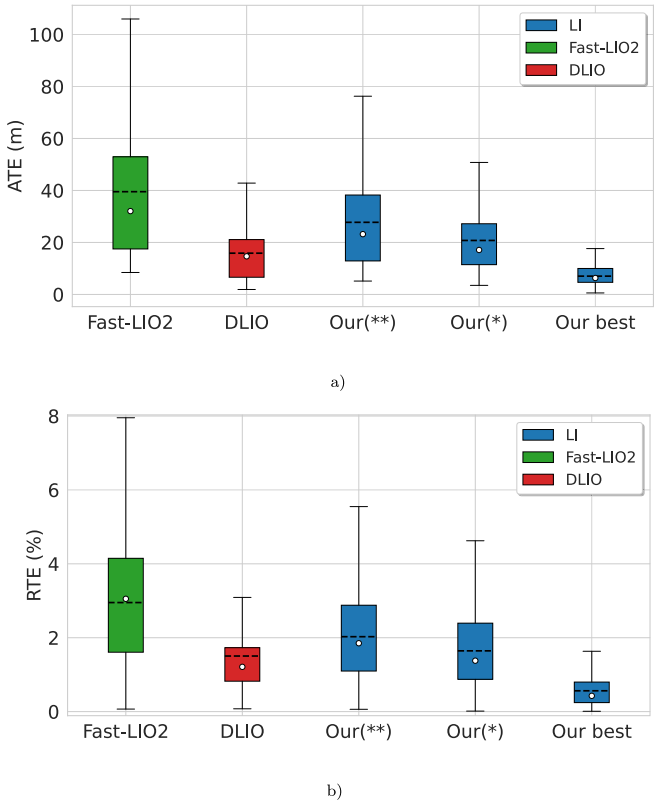


Fig. 11. Box plots showing the trajectory errors for our ablation study: (a) absolute trajectory error (ATE), (b) relative trajectory error (RTE) per 100 m. The dashed black line indicates the mean value in each configuration, while the white circle shows the median. The plot summarises the trajectory errors for 3D LiDAR and IMU in 'LI' configuration for all 3 ablation study cases (Our(**), Our(*), and Our best). No prior ALS map or GNSS-INS was used in these ablation study experiments, the system operates purely in a LiDAR-inertial configuration.

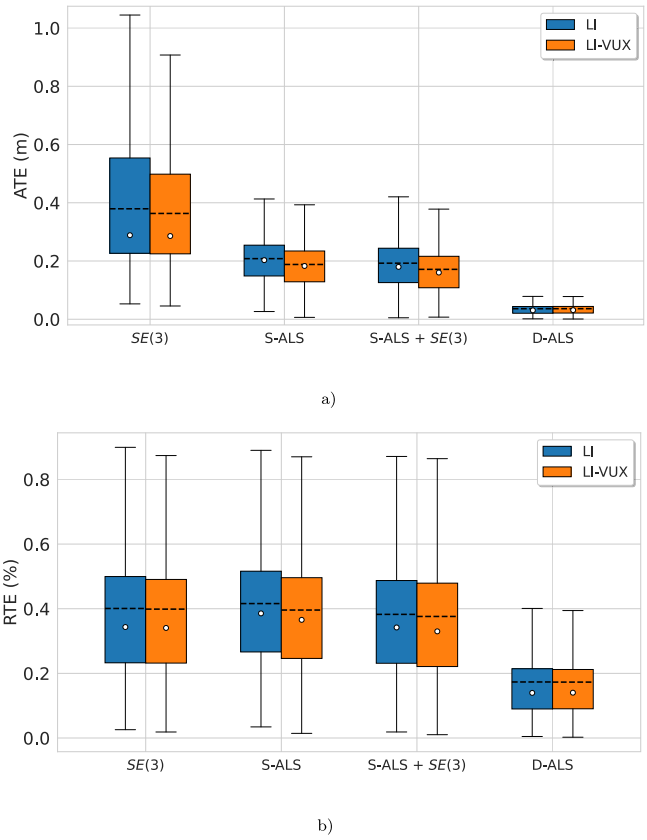


Fig. 12. Box plots illustrating trajectory errors from the ablation study assessing the contribution of the 2D VUX sensor within the full system configuration. Results are shown for configurations with and without the VUX sensor. (a) Absolute Trajectory Error (ATE) and (b) Relative Trajectory Error (RTE) per 100 m are reported. The dashed black line indicates the mean, while the white circle marks the median.

the VUX sensor remains valuable for mapping quality, as demonstrated in Figs. 8 and 9, where it contributes to increased environmental detail and improved structural representation.

An additional ablation study investigates the effect of removing the proposed Hessian and gradient scaling described in Eq. (39). The results are shown in Fig. 13. In this experiment, we evaluate the fusion of the 'LI-VUX' configuration with ALS and $SE(3)$ measurements. When the proposed scaling is omitted, the system exhibits divergence in the absolute scale. In contrast, applying the scaling significantly mitigates drift and stabilises the solution. As shown in Eq. (39), this behaviour arises because the Hessians and gradients associated with different sensor modalities have substantially different magnitudes as they are based on a different number of measurements. Without scaling, the optimisation becomes biased toward the sensor providing a larger number of measurements, thereby dominating the solution. Scaling the Hessians and gradients according to the number of measurements balances the contributions of each sensor and motivates the proposed data-fusion strategy.

The final ablation study compares the proposed tightly coupled fusion of the LiDAR-inertial system with ALS ('LI-VUX + S-ALS') versus a loosely coupled variant ('LI-VUX + S-ALS (l-coupled)'). The results are presented in Fig. 14. The key difference between the two configurations lies in the fusion strategy: in the loosely coupled approach, scan registration is performed iteratively, first against the MLS map and then against the ALS map, whereas the tightly coupled approach jointly registers each scan to both maps within a single optimisation. Both approaches achieve comparable absolute accuracy, with absolute errors of 0.18 m. The relative error is 0.39% for the tightly coupled approach and 0.6% for the loosely coupled one, indicating that although the loosely coupled method is marginally less accurate, it still provides acceptable results. Additionally, Fig. 15 demonstrates that the tightly coupled approach is nearly twice as fast, benefiting from a single joint registration instead of two sequential registrations as in the loosely coupled method.

5.4. General discussion

The results demonstrate the effectiveness of the proposed multi-sensor fusion framework and its main components. The progressive integration of additional sensors, including the 2D VUX LiDAR, GNSS-INS-derived $SE(3)$ measurements, and prior map information, consistently improves localisation accuracy and reduces drift. In particular, incorporating the 2D VUX LiDAR enhances both trajectory estimation and mapping quality, as also supported by the qualitative results. The ablation study highlights the importance of robust data-adaptive geometric uncertainty weighting and Hessian and gradient scaling for stable and balanced multi-sensor optimisation. Notably, the proposed framework does not require manual tuning of sensor covariance parameters. Instead, measurement weighting is determined automatically in a data-adaptive and robust manner, allowing the system to adaptively balance the contribution of different sensors.

Overall, the framework provides a general multi-sensor fusion mechanism that can incorporate $SE(3)$ measurements as either absolute

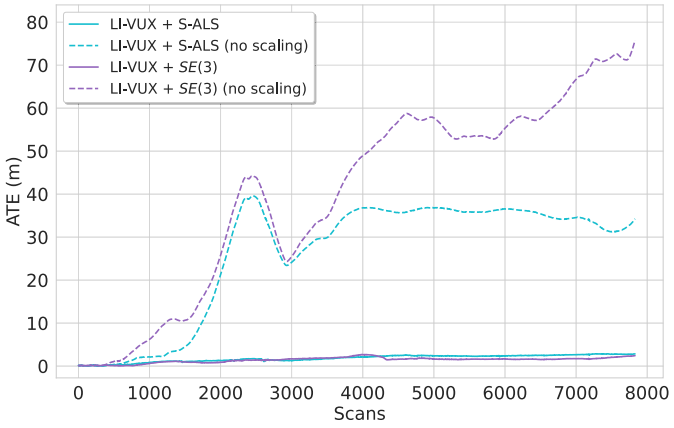


Fig. 13. The ATE plot showing the effect of no scaling of the gradients and Hessians. For a clear visualisation of the absolute error drift from the beginning of the trajectory, the trajectory alignment (used for ATE error statistics) was omitted here, instead, only the first pose of the trajectory was aligned.

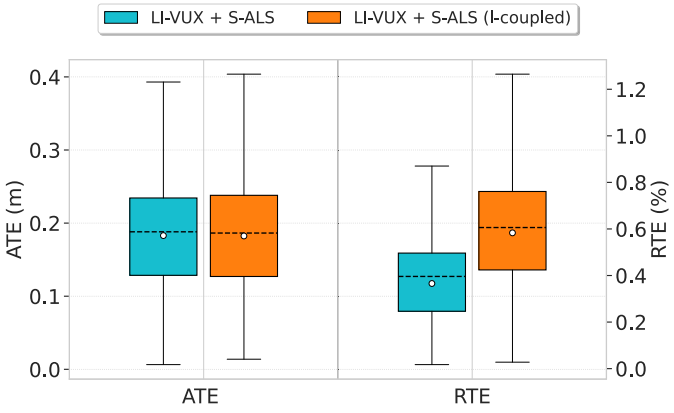


Fig. 14. Box plots of ATE and RTE for the ablation study comparing the proposed tightly coupled configuration 'LI-VUX + S-ALS' with the loosely coupled approach 'LI-VUX + S-ALS (l-coupled)'. The dashed black line denotes the mean value, while the white circle indicates the median for each configuration.

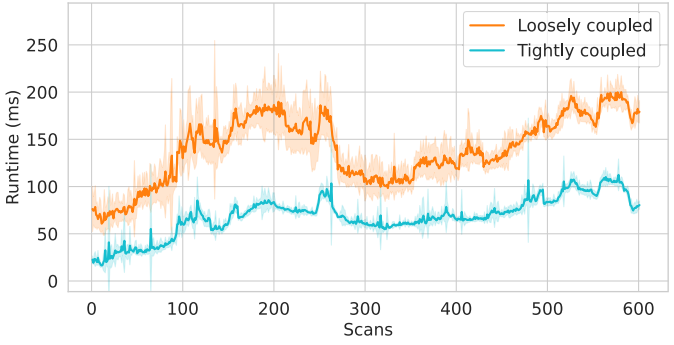


Fig. 15. Runtime comparison between the loosely ('LI-VUX + S-ALS (l-coupled)') and the proposed tightly coupled MAP-based ('LI-VUX + S-ALS') coupled methods. Each method was executed 10 times on the same dataset. Solid lines indicate the average runtime, while shaded areas represent the standard deviation (1σ).

or relative constraints, alongside LiDAR range measurements for local mapping or prior maps. While this work uses GNSS-INS-derived $SE(3)$ poses, other sources, such as visual odometry, wheel odometry, or alternative localisation systems, can be integrated in the same way. For MLS–ALS fusion, ALS data serve as a prior map, but the framework is not limited to ALS. It can incorporate any type of prior mapping, including previously acquired MLS maps. Structural discrepancies between the prior map and current observations are not explicitly modelled. Instead, nearby structures are approximated as planar constraints when present and contribute according to their estimated covariance, while recent MLS measurements dominate in areas where the prior map is less consistent. As a result, prior maps help reduce global drift, while local trajectory consistency is primarily maintained by the current MLS scans, preventing distortions caused by outdated map information. In particular, when a high-quality prior ALS map is available (e.g., D-ALS), the contribution of additional sensing modalities such as GNSS-INS-derived $SE(3)$ poses or the 2D VUX LiDAR becomes marginal for registration accuracy, as the prior map already provides sufficiently strong geometric constraints. In contrast, when the ALS prior is of lower quality or degraded (e.g., S-ALS), these additional sensors provide significant complementary information and noticeably improve both robustness and accuracy of the fusion system. Finally, a comparison of tightly and loosely coupled MLS–ALS fusion shows similar localisation accuracy, but the tightly coupled approach is nearly twice as computationally efficient and requires no user-defined parameter tuning, making it more suitable for large-scale, practical applications.

While the 2D VUX system was integrated at the point level with the 3D Hesai sensor in this work, the integration could also be performed differently. As described in the Method section (Eq. (39)), a second LiDAR can be incorporated by adding its estimated Hessian and gradient. In this alternative approach, different LiDARs could maintain separate local maps with distinct accuracies. In the current implementation, combining the 2D and 3D LiDARs into a single map also mixes their measurement uncertainties. By fusing the sensors at the Hessian and gradient level, each LiDAR could retain its own covariance, reflecting differences in sensor noise or calibration errors. This would still result in a tightly coupled system, but the coupling occurs at the Hessian/gradient level rather than at individual points. While the current point-level integration is effective, this alternative strategy remains to be explored in future work.

5.4.1. System limitations

While the presented results demonstrate the effectiveness of the proposed framework, several limitations should be acknowledged. In the current implementation, the system does not estimate or compensate for sensor time delays or desynchronisation, and such effects have not been explicitly evaluated. We therefore assume that all sensor data are time-synchronised and extrinsically calibrated, which is a standard but critical requirement in multi-sensor fusion systems. Small temporal misalignments between IMU and LiDAR can degrade motion compensation accuracy, leading to increased drift in the estimated trajectory and, in severe cases, filter inconsistency or failure. Similarly, the 2D-3D LiDAR fusion was not evaluated under asynchronous conditions. Addressing robustness to time synchronisation errors is therefore an important direction for future work, as improved temporal alignment is expected to directly benefit overall system performance.

The proposed ALS–MLS fusion is primarily geometric and does not directly depend on temporal alignment. However, this behaviour assumes that discrepancies between the prior map and current observations are locally bounded. This assumption may not hold when the prior map is outdated, incomplete, or affected by seasonal and structural changes (e.g., vegetation growth, snow cover, or construction). In such cases, the system does not explicitly detect or model these inconsistencies, and erroneous prior constraints may influence the solution and bias the resulting trajectory or map. As shown in our experiments,

higher-quality ALS priors consistently improve localisation accuracy (D-ALS), confirming the dependency of the method on prior map quality. While with lower-quality or outdated priors (e.g., 'S-ALS'), the system remains stable, but with reduced accuracy. We note that explicitly handling long-term environmental changes would require mechanisms for change detection and map updating, which is beyond the scope of this work. Such capabilities would allow the system to identify which parts of the prior map remain reliable and which should be down-weighted or ignored. However, this introduces a separate research problem of long-term map consistency and environmental change tracking, for which no suitable reference datasets were available in this study.

The proposed method has been primarily evaluated in mixed environments dominated by forests with sparse built structures. While such scenarios are challenging for LiDAR odometry due to limited geometric structure, they do not fully represent dense urban environments. We expect the method to generalise well to urban scenes with strong geometric constraints, however, this remains to be validated. Evaluation in urban scenarios and highly structured environments will require a dedicated data collection campaign, which is left for future work.

6. Conclusion

This study presented a tightly coupled multi-sensor fusion framework that integrates IMU, GNSS-INS, 2D LiDAR, 3D LiDAR, and prior ALS data for accurate online localisation and mapping. By time-synchronising and extrinsically calibrating multiple LiDAR sensors, their measurements are fused at the point level into a unified point cloud representation. Registration jointly exploits both MLS and prior ALS maps, enabling plane-based residual estimation with robust, local geometry-based uncertainty estimation. GNSS-INS and LiDAR measurements are tightly integrated within a single-state maximum a posteriori (MAP) solution, for which we derived a formulation that allows multi-sensor measurements to be combined through the accumulation of Hessians and gradients.

Experimental results over 5 km of forest road trajectories demonstrate that the proposed approach achieves high-fidelity reference accuracy. When using a dense prior ALS map, the system attains a mean absolute error of ≈ 0.04 m and a relative error of $\approx 0.17\%$. With sparse ALS priors, it achieves ≈ 0.18 m absolute error and $\approx 0.4\%$ relative error, highlighting its robustness across varying prior map modalities.

These results lead to two key observations. First, point-level fusion of heterogeneous LiDAR sensors significantly improves robustness and accuracy compared to single-sensor configurations. Second, tightly coupled integration of GNSS-INS and ALS priors provides reliable global referencing while preserving computational efficiency, making the framework suitable for real-time operation. The primary advantage of incorporating ALS as a prior lies in its ability to deliver superior absolute accuracy, enable stable online performance, and serve as a reusable, collected-once reference for repeated mapping tasks. Within the broader context of mobile mapping, this work demonstrates that tightly coupled multi-sensor fusion can effectively mitigate drift, limited FOV, and GNSS-INS degradation, while advancing MLS–ALS integration by enabling online operation, robust data-adaptive geometric uncertainty weighting based on local planarity estimated with PCA, and reduced reliance on manual tuning parameters.

The successful integration of multi-sensor fusion in a tightly coupled framework offers both stability and accuracy, opening new avenues for advancements in mapping, forestry, and autonomous navigation in a fast and cost-efficient way. Future work will explore alternative fusion strategies for 2D and 3D LiDAR at the Hessian and gradient level, address dependency on prior ALS map quality through change detection, and evaluate the framework in more diverse environments, including dense urban scenarios.

CRediT authorship contribution statement

Eugeniu Vezeteu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Aimad El Issaoui:** Writing – review & editing, Supervision, Resources, Investigation, Formal analysis, Data curation, Conceptualization. **Heikki Hyyti:** Writing – review & editing, Supervision, Data curation. **Jesse Muhojoki:** Writing – review & editing. **Eric Hyyppä:** Writing – review & editing, Supervision. **Petri Manninen:** Writing – review & editing, Investigation, Formal analysis. **Teemu Hakala:** Writing – review & editing, Software, Data curation. **Antero Kukko:** Writing – review & editing, Resources, Project administration. **Harri Kaartinen:** Writing – review & editing, Resources. **Ville Kyrki:** Writing – review & editing, Supervision. **Juha Hyyppä:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Jacobian matrices

$$\mathbf{F_x} = \begin{bmatrix} \mathbf{I} & -\frac{\Delta t^2}{2}{}^{\mathcal{M}}\mathbf{R}_I\lfloor\hat{\mathbf{a}}\rfloor_\wedge & 0 & 0 & \mathbf{I}\Delta t & 0 & 0 & 0 \\ 0 & \exp(\hat{\boldsymbol{\omega}}\Delta t) & 0 & 0 & 0 & -\mathbf{J}_r(\hat{\boldsymbol{\omega}}\Delta t)\Delta t & 0 & 0 \\ 0 & 0 & \mathbf{I} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathbf{I} & 0 & 0 & 0 & 0 \\ 0 & -{}^{\mathcal{M}}\mathbf{R}_I\lfloor\hat{\mathbf{a}}\rfloor_\wedge\Delta t & 0 & 0 & \mathbf{I} & 0 & -{}^{\mathcal{M}}\mathbf{R}_I\Delta t & -\mathbf{I}\Delta t \\ 0 & 0 & 0 & 0 & 0 & \mathbf{I} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \mathbf{I} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\mathbf{I} \end{bmatrix}$$

$$\mathbf{F_w} = \begin{bmatrix} 0 & -\frac{1}{2}{}^{\mathcal{M}}\mathbf{R}_I\Delta t & 0 & 0 \\ -\mathbf{J}_r(\hat{\boldsymbol{\omega}}\Delta t)\Delta t & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -{}^{\mathcal{M}}\mathbf{R}_I\Delta t & 0 & 0 \\ 0 & 0 & \mathbf{I}\Delta t & 0 \\ 0 & 0 & 0 & \mathbf{I}\Delta t \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{A.1}$$

where $\lfloor \mathbf{u} \rfloor_\wedge$ is the skew-symmetric matrix of vector $\mathbf{u} \in \mathbb{R}^3$ and $\mathbf{J}_r$ is the right Jacobian matrix of $SO(3)$:

$$\mathbf{J}_r(\mathbf{u}) = \mathbf{I} - \frac{1-\cos(\|\mathbf{u}\|)}{\|\mathbf{u}\|^2}\lfloor \mathbf{u} \rfloor_\wedge + \frac{\|\mathbf{u}\| - \sin(\|\mathbf{u}\|)}{\|\mathbf{u}\|^3}\lfloor \mathbf{u} \rfloor_\wedge^2. \tag{A.2}$$

The process noise covariance matrix $\mathcal{Q}$ is defined from the IMU measurement noise and bias terms contained in $\mathbf{w}$ as:

$$\mathcal{Q} = \begin{bmatrix} \mathbf{n}_\omega \mathbf{I} & 0 & 0 & 0 \\ 0 & \mathbf{n}_a \mathbf{I} & 0 & 0 \\ 0 & 0 & \mathbf{n}_{b_\omega} \mathbf{I} & 0 \\ 0 & 0 & 0 & \mathbf{n}_{b_a} \mathbf{I} \end{bmatrix}. \tag{A.3}$$

The Jacobian matrix of the LiDAR measurement model is given by:

$$\mathbf{J}_{\mathcal{L}_i} = \frac{\partial \mathbf{h}_{\mathcal{L}}(\hat{\mathbf{x}}_k \boxplus \delta \mathbf{x}_k, \pi_i)}{\partial \delta \mathbf{x}_k}\Big|_{\delta \mathbf{x}_k = \mathbf{0}} = \begin{bmatrix} {}^{\mathcal{M}}\boldsymbol{\nu}_i^\top & (\lfloor {}^{\mathcal{M}}\mathbf{s}_i \rfloor_\wedge \mathbf{G})^\top & \mathbf{U}^\top & \mathbf{G}^\top & \mathbf{0}_{1\times 12} \end{bmatrix} \in \mathbb{R}^{1\times 24} \tag{A.4}$$

where $\mathbf{G} = {}^{\mathcal{M}}\mathbf{R}_{\mathcal{I}}^{\top}{}^{\mathcal{M}}\boldsymbol{\nu}$ is the normal vector expressed at the IMU origin and $\mathbf{U} = \lfloor {}^{\mathcal{L}}\mathbf{s}_i \rfloor_\wedge {}^{\mathcal{I}}\mathbf{R}_{\mathcal{L}}^{\top}\mathbf{G}$.

We define the prior residual $\mathbf{e}$:

$$\mathbf{e}(\mathbf{x}) = \mathbf{x} \boxminus \hat{\mathbf{x}} = \log(\hat{\mathbf{R}}^\top \mathbf{R}). \tag{A.5}$$

Applying the right side perturbation leads to:

$$\mathbf{e}(\mathbf{x} \boxplus \boldsymbol{\delta}\mathbf{x}) \approx \log\Big(\hat{\mathbf{R}}^\top \mathbf{R} \exp(\delta\theta)\Big) = \log\big(\mathbf{R}_\mathbf{e} \exp(\delta\theta)\big) \tag{A.6}$$

where $\delta\theta$ is the rotation part from $\boldsymbol{\delta}\mathbf{x}$ and:

$$\mathbf{R}_\mathbf{e} = \hat{\mathbf{R}}^\top \mathbf{R}, \quad \phi = \log(\mathbf{R}_\mathbf{e}). \tag{A.7}$$

Using the first-order identity for small ϵ, we have:

$$\log(\mathbf{R}\exp(\epsilon)) \approx \log(\mathbf{R}) + \mathbf{J}_\mathbf{r}^{-1}(\phi)\,\epsilon \tag{A.8}$$

where $\mathbf{J}_\mathbf{r}^{-1}$ is the inverse **right Jacobian** of $SO(3)$ with:

$$\mathbf{e}(\delta\boldsymbol{\theta}) \approx \phi + J_r^{-1}(\phi)\,\delta\theta. \tag{A.9}$$

Hence, the prior residual Jacobian is given by:

$$\mathbf{E} = \frac{\partial}{\partial \delta \mathbf{x}}\Big[(\mathbf{x}_k^{(j)} \boxplus \delta\mathbf{x}) \boxminus \mathbf{x}_k^{(0)}\Big]_{\delta\mathbf{x}=0} = \begin{bmatrix} \mathbf{I} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \mathbf{J}_r^{-1}(\mathbf{u}) & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \mathbf{I} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathbf{I} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathbf{I} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \mathbf{I} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \mathbf{I} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \mathbf{I} \end{bmatrix} \in \mathbb{R}^{24\times 24}, \tag{A.10}$$

where $\mathbf{u}$ is the rotation part of the prior residual $\mathbf{e}$.

For the flipped residual $\mathbf{e}$ we have:

$$\mathbf{e}(\mathbf{x}) = \hat{\mathbf{x}} \boxminus \mathbf{x} = \log(\mathbf{R}^\top \hat{\mathbf{R}}). \tag{A.11}$$

Applying the right side perturbation leads to:

$$\begin{aligned} \mathbf{e}(\mathbf{x} \boxplus \boldsymbol{\delta}\mathbf{x}) &\approx \log\Big((\mathbf{R}\exp(\delta\theta))^{-1}\hat{\mathbf{R}}\Big) \\ &= \log\Big(\exp(-\delta\theta)\mathbf{R}^{-1}\hat{\mathbf{R}}\Big) \end{aligned} \tag{A.12}$$

where $\delta\theta$ is the rotation part from $\boldsymbol{\delta}\mathbf{x}$ and:

$$\mathbf{R}_\mathbf{e} = \mathbf{R}^{-1}\hat{\mathbf{R}}, \quad \phi = \log(\mathbf{R}_\mathbf{e}). \tag{A.13}$$

Using the first-order identity for small ϵ, we have:

$$\log(\exp(\epsilon)\mathbf{R}) \approx \log(\mathbf{R}) + \mathbf{J}_\mathbf{l}^{-1}(\phi)\,\epsilon \tag{A.14}$$

where $\mathbf{J}_\mathbf{l}^{-1}$ is the inverse **left Jacobian** of $SO(3)$. Here $\epsilon = -\delta\theta$, and the rotation part from $\mathbf{E}$ is $-\mathbf{J}_\mathbf{l}(\mathbf{u})^{-1}$.

The Jacobian for absolute $SE(3)$ measurement is given by:

$$\mathbf{J}_\mathcal{G}^{\text{abs}} = \frac{\partial \mathbf{h}_\mathcal{G}^{\text{abs}}(\hat{\mathbf{x}}_k \boxplus \delta\mathbf{x}_k)}{\partial \delta \mathbf{x}_k}\Big|_{\delta\mathbf{x}_k=\mathbf{0}} = \begin{bmatrix} \mathbf{I} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 18} \\ \mathbf{0}_{3\times 3} & \mathbf{I} & \mathbf{0}_{3\times 18} \end{bmatrix} \in \mathbb{R}^{6\times 24}. \tag{A.15}$$

where $\mathbf{u}^{\text{abs}}$ is the rotation part of the residual $\mathbf{r}_\mathcal{G}^{\text{abs}}$, and for relative $SE(3)$ is:

$$\mathbf{J}_\mathcal{G}^{\text{rel}} = \frac{\partial \mathbf{h}_\mathcal{G}^{\text{rel}}(\hat{\mathbf{x}}_k \boxplus \delta\mathbf{x}_k)}{\partial \delta \mathbf{x}_k}\Big|_{\delta\mathbf{x}_k=\mathbf{0}} = \begin{bmatrix} \mathbf{I} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 18} \\ \mathbf{0}_{3\times 3} & \mathbf{I} & \mathbf{0}_{3\times 18} \end{bmatrix} \in \mathbb{R}^{6\times 24}, \tag{A.16}$$

where $\mathbf{u}^{\text{rel}}$ is the rotation part of the residual $\mathbf{r}_\mathcal{G}^{\text{rel}}$.

Appendix B. Equivalence of Kalman gain forms

From Eq. (17), we show that two forms of the Kalman gain are equivalent:

$$\mathbf{K} = \mathbf{P}\mathbf{J}^\top \underbrace{(\mathbf{J}\mathbf{P}\mathbf{J}^\top + \boldsymbol{\mathcal{R}})}_{\mathcal{S}}{}^{-1} \quad \text{and} \quad \mathbf{K} = \underbrace{(\mathbf{P}^{-1} + \mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}\mathbf{J})}_{\mathcal{A}}{}^{-1}\mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1} \tag{B.1}$$

We start from the first form and apply the **Woodbury matrix identity**:

$$(\mathbf{A} + \mathbf{U}\mathbf{C}\mathbf{V})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{V}\mathbf{A}^{-1}\mathbf{U})^{-1}\mathbf{V}\mathbf{A}^{-1}, \tag{B.2}$$

with

$$\mathbf{A} = \boldsymbol{\mathcal{R}}, \quad \mathbf{U} = \mathbf{J}, \quad \mathbf{C} = \mathbf{P}, \quad \mathbf{V} = \mathbf{J}^\top. \tag{B.3}$$

Then the inverse of the innovation covariance is given by:

$$\mathcal{S}^{-1} = (\mathbf{J}\mathbf{P}\mathbf{J}^\top + \boldsymbol{\mathcal{R}})^{-1} = \boldsymbol{\mathcal{R}}^{-1} - \boldsymbol{\mathcal{R}}^{-1}\mathbf{J}(\mathbf{P}^{-1} + \mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}\mathbf{J})^{-1}\mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}. \tag{B.4}$$

Multiply by $\mathbf{P}\mathbf{J}^\top$ to get the $\mathbf{K}$:

$$\mathbf{K} = \mathbf{P}\mathbf{J}^\top(\mathbf{J}\hat{\mathbf{P}}\mathbf{J}^\top + \boldsymbol{\mathcal{R}})^{-1} \tag{B.5}$$

$$= \mathbf{P}\mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1} - \mathbf{P}\mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}\mathbf{J}(\hat{\mathbf{P}}^{-1} + \mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}\mathbf{J})^{-1}\mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1} \tag{B.6}$$

$$= \underbrace{\Big[\mathbf{P} - \mathbf{P}\mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}\mathbf{J}(\mathbf{P}^{-1} + \mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}\mathbf{J})^{-1}\Big]}_{L}\mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}. \tag{B.7}$$

Using $\mathcal{A}^{-1} = \mathbf{P}^{-1} + \mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}\mathbf{J}$, we check:

$$\mathbf{P} - \mathbf{P}\mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}\mathbf{J}\mathcal{A}^{-1} = \mathcal{A}^{-1}. \tag{B.8}$$

Multiply both sides with $\mathcal{A}$:

$$(\mathbf{P} - \mathbf{P}\mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}\mathbf{J}\mathcal{A}^{-1})\mathcal{A} = \mathbf{P}\mathcal{A} - \mathbf{P}\mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}\mathbf{J}\mathcal{A}^{-1}\mathcal{A} \tag{B.9}$$

$$= \mathbf{P}(\mathbf{P}^{-1} + \mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}\mathbf{J}) - \mathbf{P}\mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}\mathbf{J} \tag{B.10}$$

$$= \mathbf{I} + \mathbf{P}\mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}\mathbf{J} - \mathbf{P}\mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}\mathbf{J} \tag{B.11}$$

$$= \mathbf{I}. \tag{B.12}$$

Hence,

$$\mathbf{P} - \mathbf{P}\mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}\mathbf{J}(\mathbf{P}^{-1} + \mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}\mathbf{J})^{-1} = (\mathbf{P}^{-1} + \mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}\mathbf{J})^{-1}, \tag{B.13}$$

and therefore

$$\mathbf{K} = (\mathbf{P}^{-1} + \mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}\mathbf{J})^{-1}\mathbf{J}^\top\boldsymbol{\mathcal{R}}^{-1}. \tag{B.14}$$

Appendix C. Extra images

See Fig. C.16.

Data availability

The authors do not have permission to share data.

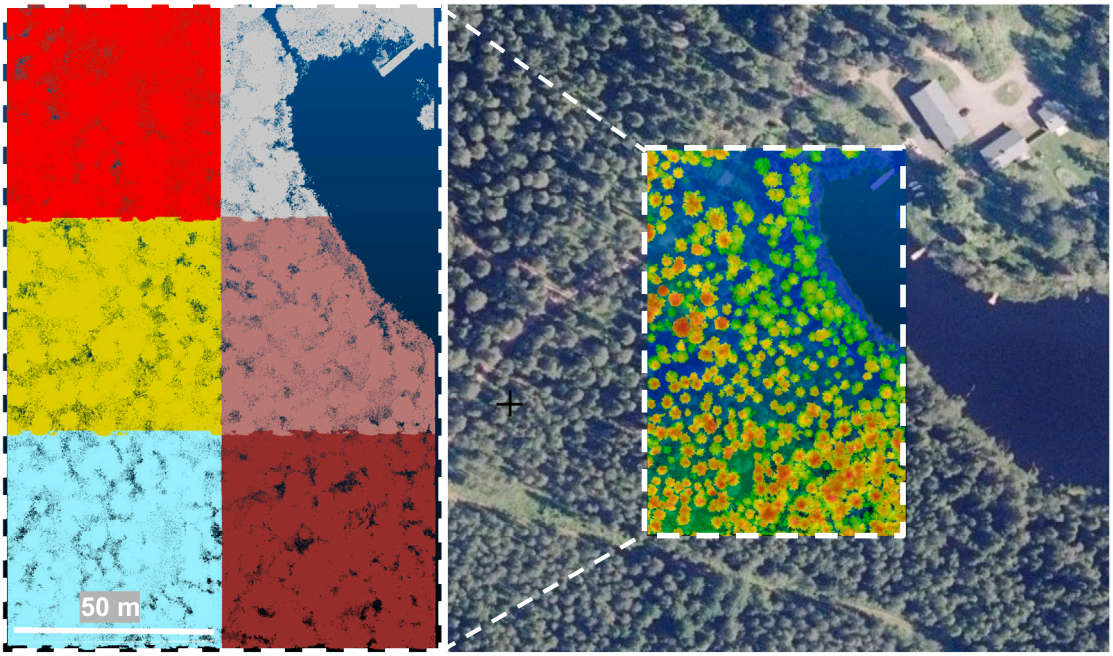


Fig. C.16. Satellite image view of the dense ALS dataset partitioned into 50 × 50 m tiles for memory-efficient loading. Individual tiles are colour-coded to illustrate the spatial tiling strategy overlaid on satellite imagery.

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